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ABSTRACT

We numerically study the fluid–structure interaction of a free-stream flow across a hydrofoil pitching at its leading edge with superimposed traveling wave-based surface undulations. We utilize an in-house code that employs the sharp interface immersed boundary method and consider a constant pitching amplitude $\theta_0 = 5°$, a constant local amplitude-to-thickness ratio $A_L = 0.15$, and wave number $K = 20$ of surface undulation. We compare the effect of surface undulation on a pitching hydrofoil with that of a hydrofoil undergoing pure pitching or experiencing pure surface undulation. The findings reveal that surface undulation on the pitching hydrofoil increases thrust on the hydrofoil. The onset of asymmetry in the vortex street occurs at a lower pitching Strouhal number (St) due to the early formation of a vortex dipole. In addition to the presence of an asymmetric inverse von Kármán vortex street, higher pitching frequencies reveal re-deflection of the asymmetric inverse von Kármán vortices. We quantified dynamics of vortex dipole to explain the occurrence of asymmetric and re-deflected reverse von Kármán vortex street. Furthermore, the analysis reveals an optimum combination of St and phase speed that yields higher propulsive efficiency, as both motions compete in generating thrust. A linearly superimposed scaling analysis for the time-averaged thrust of the combined motion is also presented. The computations and scaling are found to be in good agreement.

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I. INTRODUCTION

The study of bioinspired flows has sparked significant interest among researchers due to its relevance in designing underwater vehicles.[1–3] Fishes actively bend their bodies, tails, or fins to generate thrust and lift. The thrust generation through oscillations of the moving body parts in aquatic animals is very common. The flow around an isolated airfoil performing different motions has given several useful insights for advancement in the propulsive techniques for underwater vehicles. Fishes employ a variety of motions, such as pitching,[4–6] heaving,[7–9] body undulation,[10–12] and surface undulation,[13–16] to manipulate the surrounding flow. Notably, two distinct undulatory motions, namely, body undulation and surface undulation, have been identified for generating thrust. In body undulation, an in-phase traveling wave propagates throughout the entire body,[17,18] while in surface undulation, an anti-phase traveling wave traverses along the surface of the body.[14,16] This section presents the literature review on pure pitching, surface undulation, and combined motion, and later, we outline the objectives of the current study in this section.

In the context of *pitching motion*, it has been observed that the motion is used extensively in carangiform and thunniform fishes,[19] which inspired researchers to study the pitching motion more closely.[4,20–25] The thrust generation mechanism for the pitching motion of fin can be attributed to jet in the wake owing to the formation of reverse von Kármán vortices.[19] For pure pitching, the optimal thrust for an oscillating hydrofoil is produced for a specific range of Strouhal numbers (St) between 0.25 and 0.35, as presented by Triantafyllou *et al.*[26] Additionally, the study by Taylor *et al.*[1] demonstrated that birds, bats, and insects tend to converge within a similar range of St during cruising flight, suggesting that these animals adjust their cruising kinematics to optimize their pitching-related Strouhal number. Schnipper *et al.*[27] experimentally found different types of wakes ranging from 2S and 2P to a complex flow of 16 vortices in one pitching cycle hydrofoil. Mandujano and Málaga[28] also presented an almost similar wake transition behind a pitching hydrofoil. Bohl and Koochesfahani[29] showed that with increasing pitching frequency, the flow behind hydrofoil changes from von Kármán vortex street to reverse von Kármán vortex street, but the transition does not coincide exactly with the drag-to-thrust transition. Das *et al.*[22] also found that a critical St exists for a transition of drag to thrust for different Reynolds numbers (Re). They also presented a scaling analysis for a dependence of thrust and propulsive efficiency on Re and St.

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Floryan *et al.*[30] formulated a scaling law for hydrofoils undergoing a pure pitching motion while incorporating the effect of added mass forces. Notably, they observed a similar scaling trend in time-averaged thrust coefficient in that $\langle C_T \rangle$ scales as St^2 as previously documented,[22,28,31] provided that the pitching amplitude and drag on the hydrofoil remained within small limits. Das *et al.*[32] showed that for a self-propelled pitching motion, which is unidirectional or bidirectional free, the scaling law for C_T–St and St–Re is similar for both the cases. Lagopoulos *et al.*[33] showed that the universal length scale to characterize the drag-to-thrust transition is the average trajectory length covered by the entire hydrofoil in one time period. They argued that the Strouhal number calculated concerning this length scale converges toward unity during the drag-to-thrust transition.

Apart from these sinusoidal pitching studies, researchers have delved into the propulsive performance of non-sinusoidal pitching motion as well.[24,34–39] These investigations unveiled a variety of vortical structures that are attainable by changing the waveform in non-sinusoidal pitching. A single strong vortex forms in the fast part of the cycle, while multiple same-sign vortices shed in the slow part of the non-sinusoidal pitching cycle.[34] The flexibility of pitching hydrofoil has also been an intriguing topic for many researchers.[20,40–43]

In the context of *surface undulation*, Essapian[44] first observed the presence of waves or wrinkles on the skin of a fast-moving dolphin during prey capture, which inspired researchers to investigate the traveling wave-based surface undulation in the propulsion of fishes. In the research of Wu *et al.*[45] and Xu *et al.*,[46] it was observed that traveling waves on the leeward side of a cylinder effectively suppress the von Kármán vortex street that occurs behind the cylinder. Wu *et al.*[45] investigated the effectiveness of using traveling waves on the leeward side to reduce drag on bluff bodies. The authors found a significant reduction of drag (up to 85%). Xu *et al.*[46] studied the effect of altering the direction of waves on both the upper and lower surfaces at the rear of the cylinder. They concluded that the downstream moving traveling waves on both the surfaces resulted in a more significant drag reduction compared to an upstream wave motion. Thompson and Goza[47] studied the effects of traveling wave-based surface undulation on the lift at low and stalled angles of attack. They showed that the changes in streamlined curvature due to surface undulation can lead to an increased lift through alterations in the pressure distribution on the suction side of the hydrofoil. In the present study, surface undulation is considered on both the hydrofoil surfaces to investigate its impact on the thrust/drag force, as opposed to the lift force. The most significant lift benefits of undulation are observed in the lock-on region, where the surface undulation frequency matches the vortex street frequency of an unactuated hydrofoil, particularly at stalled angles of attack.

Tian *et al.*[48] numerically analyzed the self-propelled motion of a hydrofoil with a traveling wave superimposed on it. They varied the forward velocity of the hydrofoil with different wave parameters and found that the mean forward velocity of the hydrofoil increases with amplitude as well as the speed and wave number of the traveling wave. In a follow-up work, Tian *et al.*[49] compared the final mean cruising velocity of self-propelled undulation motion with the traveling wave surface motion, which are 0.58 and 0.52 times the wave speed, respectively. Sooraj and Agrawal[50] found that corrugated hydrofoil behaves as a normal hydrofoil having a surface undulation as the vorticity forms in valleys of the corrugated hydrofoil, making corrugated hydrofoil more efficient than normal hydrofoil. The vortices trapped in the valleys are defined as "fluid roller bearing," and these trapped vortices in the trough of the wave serve as a roller bearing, which reduces the drag on the body.[51] Chuijie *et al.*[51] focused on the delay of stall using a non-uniform amplitude traveling wave on the upper surface of the hydrofoil. Akbarzadeh and Borazjani[13] also confirmed the delay in the stall of the inclined hydrofoil using a traveling wave on its upper surface at $Re = 5 \times 10^4$. Albers and Schröder[52] employed large eddy simulation to study flow over a NACA4412 hydrofoil with a sinusoidally actuated suction side surface at $Re = 4 \times 10^4$ for the case of highest drag reduction. They found that by introducing surface actuation, the flow separation point can be delayed, and the size of the turbulent wake behind the airfoil can be reduced, leading to an increased lift and a reduced drag. The turbulent kinetic energy reduces on the suction side of the airfoil in comparison to the non-actuated airfoil. As the turbulent wake shortens, its flow remains attached to the airfoil, thereby delaying the stall and increasing the lift coefficient. He *et al.*[53] conducted an experimental study to compare the aerodynamic characteristics of a rigid hydrofoil with that of a passively and actively actuated suction side hydrofoil at various angles of attack. Their investigation revealed that the actively actuated suction side of the hydrofoil exhibited a maximum lift-to-drag ratio after the stall angle, primarily due to the delayed stall induced by the periodic vibration of a portion of the suction side of the hydrofoil.

In the context of *combined motion*, pitching and heaving motions have been studied extensively.[54–64] In an experimental study, Anderson *et al.*[65] found that the optimal thrust generation occurs when the Strouhal number lies between 0.25 and 0.4, pitching is pivoted at 1/3 of the hydrofoil chord length, and the phase difference between the pitching and heaving is about 75°. They further achieved propulsive efficiency as high as 87% in their study. Andersen *et al.*[57] identified the wake patterns and correlated them with the drag-to-thrust transition in a pitching hydrofoil. They highlighted that the thrust generation is through a meandering jet behind the hydrofoil, stemming from the creation of two counter-rotating trailing edge vortices in one pitching cycle, known as the reverse von Kármán vortex street. Apart from the pitching- and heaving-based combined motion, Alben[66] studied the combined motion based on pitching, heaving, and shifting motion on a flexible plate. Quinn *et al.*[67] found that the incorporation of pitching motion on a heaving motion of a flexible panel enhances the efficiency by a factor of 1.7. Additionally, they also found that the optimal St range reduces from 0.4 to 0.53 for pure heaving motion to 0.26–0.33 for the combined heaving with pitching motion. A similar result is observed by Van Buren *et al.*[68] for a rigid hydrofoil, where adding heaving motion to pitching motion with an incremental increase in heave amplitude increases the maximum efficiency by almost the same factor in comparison to the pure pitching motion.

The above-mentioned literature review indicates a gap in understanding the effect of surface undulation on a pitching hydrofoil as a mechanism for the propulsion of bioinspired underwater vehicles. Despite extensive research on combined pitching and heaving motions, the effect of surface undulation on a pitching hydrofoil remains unexplored. In our previous work,[14] we found that the thrust generated by the surface undulation is due to the pressure-suction mechanism—the same as that for the pitching motion. We hypothesize here that the combined motion of surface undulation and pitching will increase the thrust force because of jet formation in the wake due to the combined effect of two types of motion.

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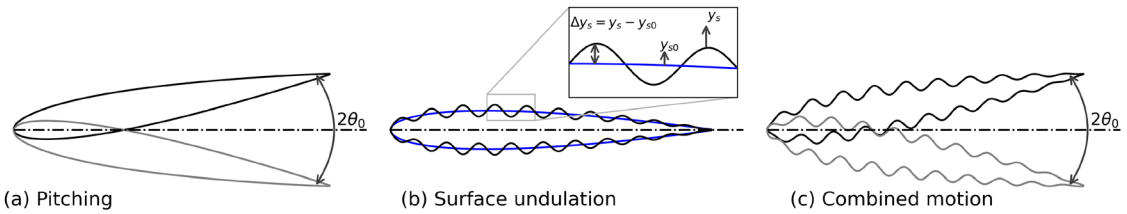


FIG. 1. Schematic diagram for (a) pure pitching, (b) pure surface undulation, and (c) combined motion of surface undulation on pitching hydrofoil. Zoomed-in view shows y_{s0} as the surface without surface undulation represented through a blue line and y_s as the surface with surface undulation represented by the black curve.

The primary objective of this study is to analyze the impact of surface undulation on the flow characteristics and propulsive performance of a pitching hydrofoil. The study aims to address the following research questions: (i) How does surface undulation affect the fluid dynamics around a pitching hydrofoil? (ii) What is the spatial distribution of thrust along the time-varying wavy surface? (iii) How does the introduction of surface undulation influence the lift variation in a pitching hydrofoil?

To address the above-mentioned inquiries, we conduct a numerical investigation of flow over a sinusoidal leading-edge pitching of NACA0012 hydrofoil with surface undulation motion on the surface using an in-house developed fluid–structure solver. The rest of the paper is organized as follows: Sec. II presents details regarding the problem statement and outlines the numerical methodology employed in the present study. Section III provides a comprehensive presentation of the results, encompassing flow dynamics and propulsive parameters. Finally, Sec. IV presents the conclusions derived from the outcomes of this numerical investigation.

II. PROBLEM STATEMENT AND NUMERICAL DETAILS

A. Kinematics of the hydrofoil

In the present problem, we focus on the hydrodynamic effect of a traveling wave-based surface undulations on a pitching hydrofoil in a free stream flow. Figure 1 shows a schematic of pure pitching, pure surface undulation, and the combined motion of pitching with surface undulation motion. We consider traveling waves on the top and bottom surfaces for surface undulation in antiphase, which moves in the streamwise direction for the pure surface undulation and combined motion, as shown in Fig. 1.

The schematic for the non-dimensional one-way coupled fluid–structure interaction computational setup is shown in Fig. 2. The figure shows that free stream fluid flows across a hydrofoil subjected to pitching motion around its leading edge, along with a streamwise traveling wave-based surface undulation applied on the hydrofoil's top and bottom surfaces. Both suction (top) and pressure (bottom) surfaces are actuated for surface undulations and are in anti-phase. The amplitude $a(x)$ of surface undulation is zero at the leading and trailing edges with an *increasing–decreasing* variation in the streamwise direction similar to the problem considered earlier in Shukla *et al.*[14] The *kinematics* of the sinusoidal pitching of the hydrofoil is given as

$$\theta(t) = \theta_0 \sin(2\pi f_p t), \tag{1}$$

where $\theta(t)$ depicts the instantaneous variation in the amplitude of the pitching motion, and θ_0 and f_p denote the pitching amplitude and the pitching frequency, respectively. The symbols used in the present study are defined in Nomenclature. Assuming that the pitching amplitude is significantly smaller than the chord length, the hydrofoil is subjected to both the pitching motion and a streamwise traveling wave-based surface undulation. The *kinematic model* for the streamwise traveling wave-based surface undulation is

$$y_s(x,t) = y_{s0}(x) + a(x)\sin[2\pi(kx - ft)], \tag{2}$$

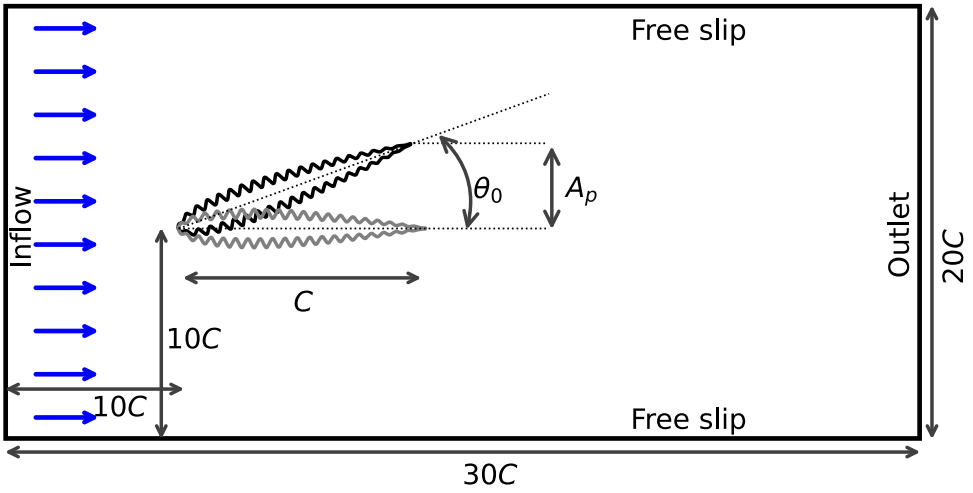


FIG. 2. Schematic for the computational setup for the combined motion of traveling wave-based surface undulation and pitching motion. θ_0 represents the maximum angular displacement due to the pitching motion.

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where y_{s0} and $a(x)$ are the y-coordinate of the undeformed hydrofoil surface and the chordwise varying amplitude of the wave, respectively, and the traveling wave-based surface undulation has a wave number k and frequency f. The surface undulation amplitude $a(x)$ follows an increasing–decreasing trend, as shown in Fig. 2. The streamwise variation in amplitude $a(x)$ is directly proportional to the local hydrofoil thickness ($2\,y_{s0}$). Therefore, the chordwise varied amplitude $a(x)$ is given as

$$a(x) = 2A_L y_{s0}, \tag{3}$$

where A_L is a proportionality constant, named as the local amplitude-to-thickness ratio of the surface undulation. The non-dimensional form of the combined motion, employing the chord length C as the length scale and the free-stream velocity U_∞ as the velocity scale, is as follows:

$$\begin{bmatrix} X_c \\ Y_c \end{bmatrix} = \begin{bmatrix} X_h \\ Y_h \end{bmatrix} + \underbrace{\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}_{\text{Pitching motion}} \underbrace{\begin{bmatrix} X_{s0} \\ Y_{s0}(x) + A(x)\sin[2\pi K(X - c^*\tau)] \end{bmatrix}}_{\text{Surface undulation}}, \tag{4}$$

where the instantaneous pitching angle θ is given in its non-dimensional form as

$$\theta = \theta_0 \sin\left(\pi \frac{St}{A_p}\tau\right). \tag{5}$$

Further, $Y_s = y_s/C$; $Y_{s0} = y_{s0}/C$; $X_{s0} = x_{s0}/C$; $X_h = x_h/C$; $Y_h = y_h/C$; and $\tau = U_\infty t/C$. Here, (x_h, y_h) represents the position of the hydrofoil's leading edge, while (x_{s0}, y_{s0}) denotes the profile of the undeformed hydrofoil within the domain. Equation (4) shows the deformation on the hydrofoil due to the surface undulation and pitching motions. The effects of combined motion on the hydrodynamics around the hydrofoil are studied by defining the non-dimensional parameters for pitching and surface undulation separately. The non-dimensional mean to peak amplitude A_p and Strouhal number St are defined for the pitching motion. The selected non-dimensional parameters for surface undulations are the local amplitude-to-thickness ratio A_L, wave number K, and non-dimensional phase speed c^*. These non-dimensional parameters are defined as follows:

$$A_p = \frac{a_p}{C}; St = \frac{2f_p a_p}{U_\infty}; A_L = \frac{a(x)}{2y_{s0}}; K = Ck; c^* = \frac{c}{U_\infty}. \tag{6}$$

We have specifically selected the Strouhal number St instead of the reduced frequency f_p^* ($=f_pC/U_\infty$) to understand the effect of pitching motion on the flow characteristics and propulsive performance. St is the ratio of the maximum pitching velocity at the trailing edge to the free stream velocity, and c^* is the ratio of phase speed of surface undulation to the free stream velocity. Therefore, St and c^* are the governing parameters for pure surface undulation and pure pitching motion, respectively. The variation of these parameters for the present work is shown in Table I.

The schematic of the computational domain and the boundary conditions (BCs) for the combined motion is shown in Fig. 2. The inlet is prescribed as a free stream velocity U_∞. Free slip BC is used at the top and bottom boundaries, and a convective BC (with a convective velocity equal to 1) is used at the outlet. The no-slip boundary condition is enforced on the surface of the hydrofoil, denoted by $U = U_c$, $V = V_c$, and $\partial P/\partial n = 0$, where n represents the normal direction to the body surface. The velocities $U = U_c$ and $V = V_c$ represent the solid surface velocity, which is obtained by differentiating Eq. (4) with respect to time, expressed as follows:

$$\begin{bmatrix} U_c \\ V_c \end{bmatrix} = \begin{bmatrix} -\sin\theta & -\cos\theta \\ \cos\theta & -\sin\theta \end{bmatrix} \begin{bmatrix} X_{s0}\dfrac{\partial\theta}{\partial\tau} + V_s \\ (Y_{s0}(x) + A(x)\sin[2\pi K(X - c^*\tau)])\dfrac{\partial\theta}{\partial\tau} \end{bmatrix}. \tag{7}$$

Here, Vs and θ' present the non-dimensional velocity in the y-direction due to pure surface undulation and angular velocity of the pure pitching motion, respectively. The surface undulation velocity is given as

$$V_s = -2\pi K c^* A_L Y_{s0}(X)\cos[2\pi K(X - c^*\tau)]. \tag{8}$$

TABLE I. Parametric details for pitching and surface undulation coupled motion.

Governing parameters	Parametric values
Pitching amplitude: θ_0	5°
Strouhal number: St	0.1–0.6 (in steps of 0.1)
Local amplitude-to-thickness ratio: A_L	0.15
Non-dimensional wave number: K	20
Non-dimensional phase speed: c^*	0–5 (in steps of 1)
Reynolds number: Re	5000

B. Numerical and parametric details

1. Numerical methodology

In the present study, we consider the flow to be unsteady, incompressible, and viscous. The governing equations for the two-dimensional flow are the continuity and Navier–Stokes equations. The present one-way coupled problem with a prescribed motion-based moving solid boundary is discretized on a finite volume method-based collocated grid and solved using a level-set-based immersed interface method. A level set function, which is a signed normal distance function, is calculated at every time step to capture the instantaneous interface between the solid and the fluid. The level set function is calculated using the minimum distance and winding algorithms governed by the prescribed combined pitching and surface undulation motions.

The present in-house solver uses a fully implicit pressure projection method on collocated grid. The QUICK scheme and the central-difference scheme are used for the advection and diffusion terms of Navier–Stokes equations, respectively. Further information on the in-house solver is available in the study of Thekkethil and Sharma.[69,70] Validation of the present in-house solver was presented for a surface undulation on the rear side of a cylinder and is given in our recent study[14] while that for pure pitching was presented in the study of Thekkethil and Sharma.[69] The validation results are not reproduced here for the sake of brevity.

2. Parametric details

Table I presents values of the governing parameters for the present problem. As discussed earlier, only two parameters, St and phase

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speed c^*, are varied while keeping the other parameters constant. The table shows that the local amplitude-to-thickness ratio and wave number of surface undulations are kept constant at 0.15 and 20, respectively. We have done a parametric study of the pure surface undulations with different values of K and found that for $K = 20$, the system is the most efficient.[16] Therefore, the value of K has been kept constant at 20 for the present problem. Note that the phase difference between the top and bottom surfaces of the hydrofoil is taken as 180° for all cases, as maximum thrust is obtained at this particular phase difference.[16]

For the present problem, a chord length-based Reynolds number $Re(= \rho U_\infty C/\mu)$ is fixed at 5000, which is under a two-dimensional flow regime for pitching case, specifically up to pitching Strouhal number $St = 0.6$ as presented in Ref. 71. Additionally, in an empirical investigation, Alam *et al.*[72] reported that the Reynolds number outlined for the present study falls within the domain of ultra-low Re values, where the flow retains its laminar attributes across a wide range of angles of attack. Subsequently, a separate inquiry conducted by Gupta *et al.*,[73] focusing on the specific case of $Re = 5 \times 10^3$ as employed in the present problem, disclosed that the flow preserves a two-dimensional character until an angle of attack $\alpha = 6°$ is reached, while a transition toward three-dimensional behavior happens at $\alpha = 7°$. As the maximum angular displacement is limited to 5°, the present simulations fall within the 2D flow regime.

The propulsive performance of a pitching hydrofoil with surface undulation is assessed using the mean thrust coefficient $\langle C_T \rangle$ and the propulsive efficiency η_p. The coefficient of thrust and lateral force coefficient are first defined as

$$C_T = \frac{F_T}{\frac{1}{2}\rho U_\infty{}^2 C}; \quad C_L = \frac{F_L}{\frac{1}{2}\rho U_\infty{}^2 C}; \tag{9}$$

where F_T and F_L represent the instantaneous thrust and lateral forces acting on the hydrofoil, respectively. The time-averaged or mean thrust coefficient is given by

$$\langle C_T \rangle = \frac{1}{T}\int_\tau^{T+\tau} C_T(\tau) d\tau. \tag{10}$$

The propulsive efficiency of the traveling wave-based surface undulation is next defined as

$$\eta_p = \frac{P_{out}}{P_{in}} = \frac{\langle F_T \rangle U_\infty}{\left\langle \int (f_L v_c - f_T u_c) ds \right\rangle} = \frac{\langle C_T \rangle}{\left\langle \int (c_L V_c - c_T U_c) dS \right\rangle} = \frac{\langle C_T \rangle}{\langle C_P \rangle}, \tag{11}$$

where f_L and f_T represent the local instantaneous lateral force and local thrust force, respectively, applied to the hydrofoil per unit surface area, and c_L and c_T are the local instantaneous coefficients of lateral force and thrust force, respectively, on the hydrofoil per unit surface area. $\langle\rangle$ denotes quantities averaged over time for 20 cycles of oscillations. Additionally, (u_c, v_c) and (U_c, V_c) represent the dimensional and non-dimensional forms of the velocity of the combined motion, respectively. $\langle C_P \rangle$ represents the input power coefficient required to maintain the prescribed combined motion. The input power coefficient calculation considers both the energy required for surface undulation motion and the power consumed in sustaining rotational pitch. Finally, ds and dS represent a dimensional and non-dimensional infinitesimal surface area of the hydrofoil, respectively.

III. RESULTS AND DISCUSSION

This section is organized as follows: we first analyze the flow fields around the hydrofoil and compare the combined motion with the pure pitching and pure surface undulation cases. In particular, this analysis helps to elucidate the correlation between the flow dynamics and the propulsive characteristics of the prescribed combined motion such as C_T, C_L, and cycle-averaged quantities ($\langle C_T \rangle$, $C_{L_{rms}}$, and η_p). Sections III A and III B deal with the flow dynamics around the hydrofoil, while Sec. III C quantifies the propulsive parameters for the given parametric range of the combined pitching and surface undulation motion.

A. Flow field around the hydrofoil

In this section, we analyze the flow dynamics around a hydrofoil subjected to the combined motion and compare it to the individual cases of pure pitching and pure surface undulation. Figure 3 shows the instantaneous vorticity contours for pure surface undulation cases, i.e., $St = 0$, at different phase speeds c^*. A simulation with stationary corrugated hydrofoil ($c^* = 0$) was also carried out. In this case, a forward vortex sheet (FV_{Sheet}) forms behind the hydrofoil. The flow inside the trough of each wave forms isolated vortices for $c^* < 1$, while at higher $c^* > 1$, the flow emanates from one surface of waves, which pushes the surrounding fluids and lands on the other surface of each hat-like wavy structure, which sucks the surrounding fluids due to lower pressure region.[14,51,74,75] As the phase speed increases, the flow separation near the trailing edge of the hydrofoil becomes evident, and the separation point gradually moves toward the leading edge, as seen in Figs. 3(b)–3(f).

In our previous paper,[16] we considered the flow behind the hydrofoil near the trailing edge, i.e., in the near wake region, to define the flow regimes in the pure surface undulation with wave number $K = 20$. In the present study, we define the near-wake regime as the area within the chord length of hydrofoil, measured from its trailing edge, while the far-wake regime extends beyond this chord length from the trailing edge. Here, for phase speeds $c^* \leq 2$, Figs. 3(b) and 3(c) reveal that the flow far downstream of the hydrofoil undergoes a transition from a FV_{Sheet} to a forward vortex street (FV_{Street}). Notice that at $c^* = 3$, two pairs of forward vortex sheets (2P-FV_{Sheet}) form behind the hydrofoil. Furthermore, as the phase speed increases further, the near wake transforms into a two-pair reverse vortex sheet (2P-RV_{Sheet}), accompanied by the formation of side vortices. These side vortices form as the flow separation shifts toward the leading edge after crossing the maximum thickness of the hydrofoil. Note that the angle of flow separation progressively increases as c^* increases. The flow separation occurring before the maximum thickness of the hydrofoil results in the formation of larger side vortices due to the roll-up of the vortex sheets, as shown in Figs. 3(e) and 3(f). The terms vortex sheet and vortex street are used in a manner consistent with our previous work.[14,16] The justification for applying the terms "vortex sheet" and "vortex street" flow regimes in the context of surface undulation motion is as follows: The continuous vortex sheet trailing the hydrofoil extends over a significant longitudinal distance, with its lateral dimension approaching the amplitude of the surface undulation wave. This is why we refer to it as a vortex sheet since it exhibits a jump in tangential velocity within an extremely thin layer in the lateral direction.

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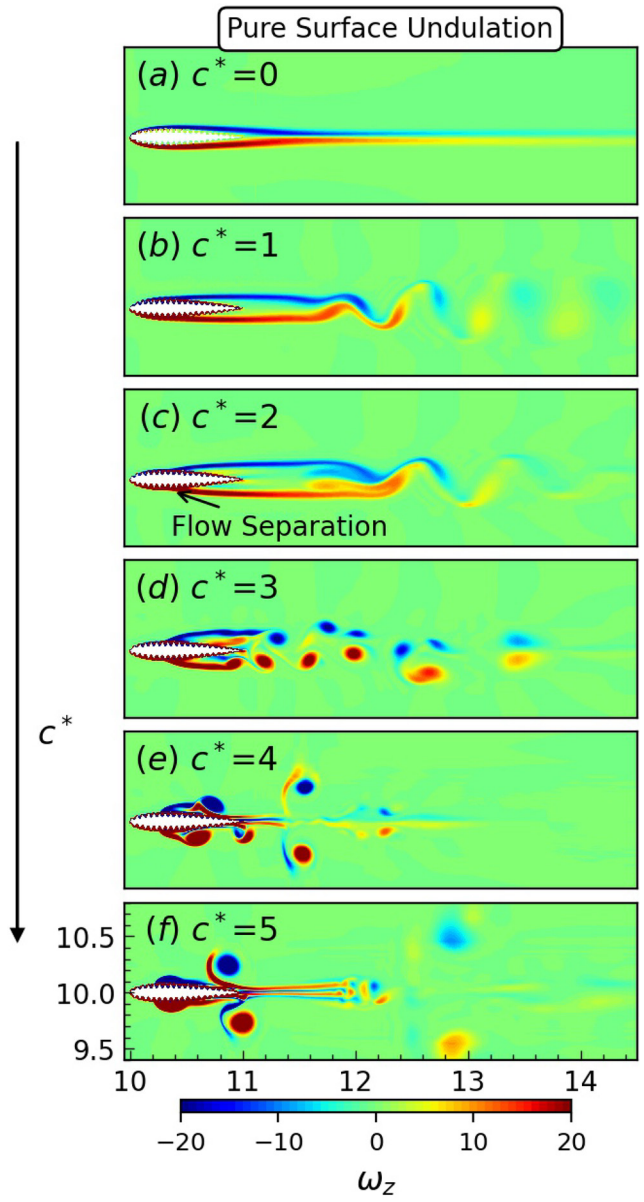


FIG. 3. Instantaneous vorticity contours behind the hydrofoil for different phase speeds c^* of the surface undulation without pitching motion.

The vorticity contours for pure pitching and combined motion with various pitching Strouhal number (St) and surface undulation phase speed (c^*) are depicted in Figs. 4–6. In the case of pure pitching with $St = 0.1$, the flow pattern behind the hydrofoil forms a forward vortex street (FV_{Street} or von Kármán vortex street), as shown in Fig. 4(a). However, for a corrugated hydrofoil characterized by sinusoidal bumps on its surface (i.e., surface undulation with $c^* = 0$), the flow changes to two pairs (2P) of opposite vortices, as shown in Fig. 4(b). The vortex trapped within the trough of the surface undulation wave, for the case of $c^* = 0$, interacts with the trailing-edge vortices generated by the pitching motion. This interaction leads to the formation of 2P vortices behind the hydrofoil. When a traveling wave-based surface undulation ($c^* > 0$) is applied to the pitching hydrofoil at $St = 0.1$, the flow behind the hydrofoil still forms 2P vortices for a lower phase speed ($c^* \leq 2$). As the phase speed c^* increases, the flow behind the pitching hydrofoil undergoes a transition from 2P vortex street to 2P + mS vortex street, where multiple single (mS) vortices emerge alongside two pairs of opposite vortices, as shown in Figs. 4(e) and 4(f). This phenomenon occurs as the vortex sheets resulting from the surface undulation are pushed by the pitching motion and subsequently roll up to form the 2P + mS vortices. The value of m may vary from 2 to 4 depending on the value of St and c^*. At $c^* = 5$, the flow behind the hydrofoil with $St = 0.1$ exhibits a non-periodic vortex structure, as illustrated in Fig. 4(g).

For the case of $St = 0.2$, the flow behind the pure pitching hydrofoil displays a reverse vortex street (RV_{Street}) or inverted von Kármán vortex street, as depicted in Fig. 4(h). This shows the existence of a critical St at which the flow behind the pitching hydrofoil transitions from von Kármán vortex street to reverse von Kármán vortex street, consistent with previous studies.[22,27,57] When a corrugated hydrofoil is used, the flow pattern remains the same as RV_{Street}, but the magnitude of the vortices increases, as shown in Fig. 4(i). Here, the magnitude of vortices means the strength of the vortices, which is calculated using the circulation. The strength of the vortices increases with the increase in the c^*. However, RV_{Street} at the phase speed $c^* = 2$ changes to a neutral wake at $c^* = 3$, as shown in Fig. 4(l). The neutral wake occurs when the vortices align along the centerline of the mean pitching position, suppressing jet formation behind the hydrofoil. Furthermore, for $c^* = 4$, a $2P + 2S$ vortex street forms behind the hydrofoil, characterized by the appearance of isolated vortices on either side of the hydrofoil due to the traveling wave-based surface undulation, as shown in Fig. 4(m). This change is attributed to the formation of an isolated vortex on either side of the hydrofoil due to the traveling wave-based surface undulation. The flow exhibits non-periodicity for $c^* = 5$ case, as shown in Fig. 4(n).

At $St = 0.3$, a reverse vortex street (RV_{Street}) is generated by the pure pitching motion, resulting in a momentum excess jet flow at the back of the hydrofoil as depicted in Fig. 5(a). The incorporation of surface undulation does not alter the vortex structures behind the hydrofoil, except for an increase in the vorticity strength up to $c^* \leq 4$. However, for pitching at $St = 0.3$, the vortex structures become non-periodic when the surface undulation phase speed reaches $c^* = 5$. The pure pitching case at $St = 0.4$ induces an asymmetric RV_{Street} that is primarily observed in the far-downstream region. As c^* increases, the asymmetricity shifts toward the hydrofoil's trailing edge and the asymmetricity angle grows, as presented in Figs. 5(h)–5(l). The non-periodic nature of the flow is displayed in Fig. 5(n) for a high surface undulation phase speed of $c^* = 5$.

The pure pitching case at $St = 0.5$ and 0.6 generates an asymmetric RV_{Street} behind the hydrofoil, as illustrated in Figs. 6(a) and 6(g). The position of the asymmetry moves toward the tail of the hydrofoil with an increase in either St or c^*, as demonstrated in Figs. 5 and 6, respectively, for $St = 0.4$ and 0.5. Furthermore, the angle of asymmetry also increases with an increase in phase speed c^*. At a phase speed of $c^* = 2$ for $St = 0.5$ and $c^* = 1$ for $St = 0.6$, the asymmetric RV_{Street} behind the hydrofoil deflects to the opposite direction of the asymmetry in the far downstream.

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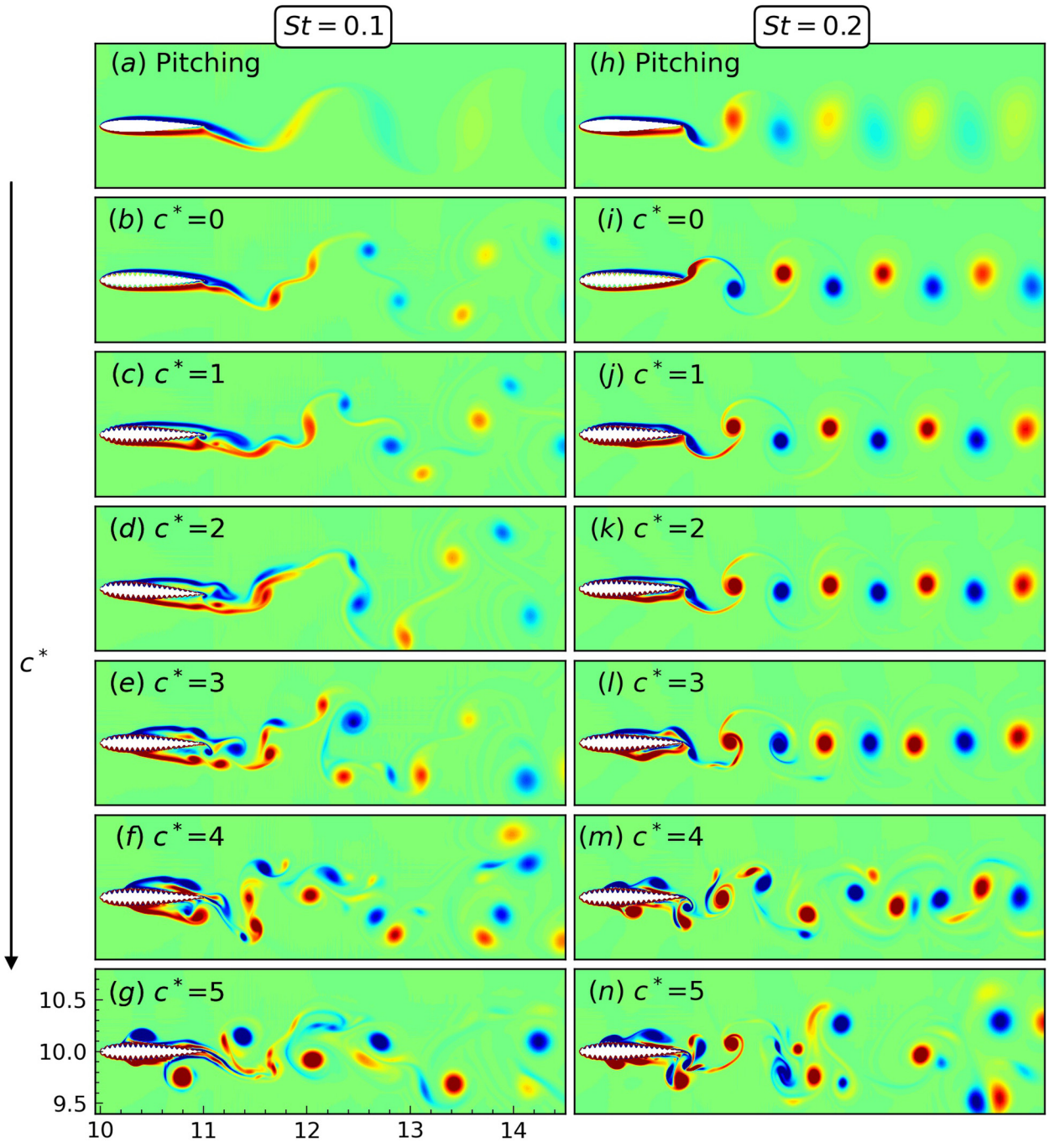


FIG. 4. Instantaneous vorticity contours for different phase speeds c^* of the surface undulation at pitching frequency (a)–(g) $St = 0.1$ and (h)–(n) $St = 0.2$. The first row shows only the pitching motion, while the other rows show a combined pitching motion with surface undulation.

Figure 7 illustrates the flow regimes for different cases involving pure pitching, pure surface undulation, and combined motion. The flow patterns downstream of the hydrofoil manifest nine discernible flow structures, namely, FV_{Sheet}, FV_{Street}, 2P-RV_{Street}, RV_{Street}, 2P, 2P + mS, 2P + 2S, asymmetric RV_{Street}, and re-deflected RV_{Street}. In addition to these flows, a non-periodic motion is also observed at high surface undulation phase speed and low Strouhal number ($St \leq 0.4$). Notably, 2P-RV_{Street} is accompanied by side vortices flanking the hydrofoil, as shown in Figs. 3(d)–3(f).

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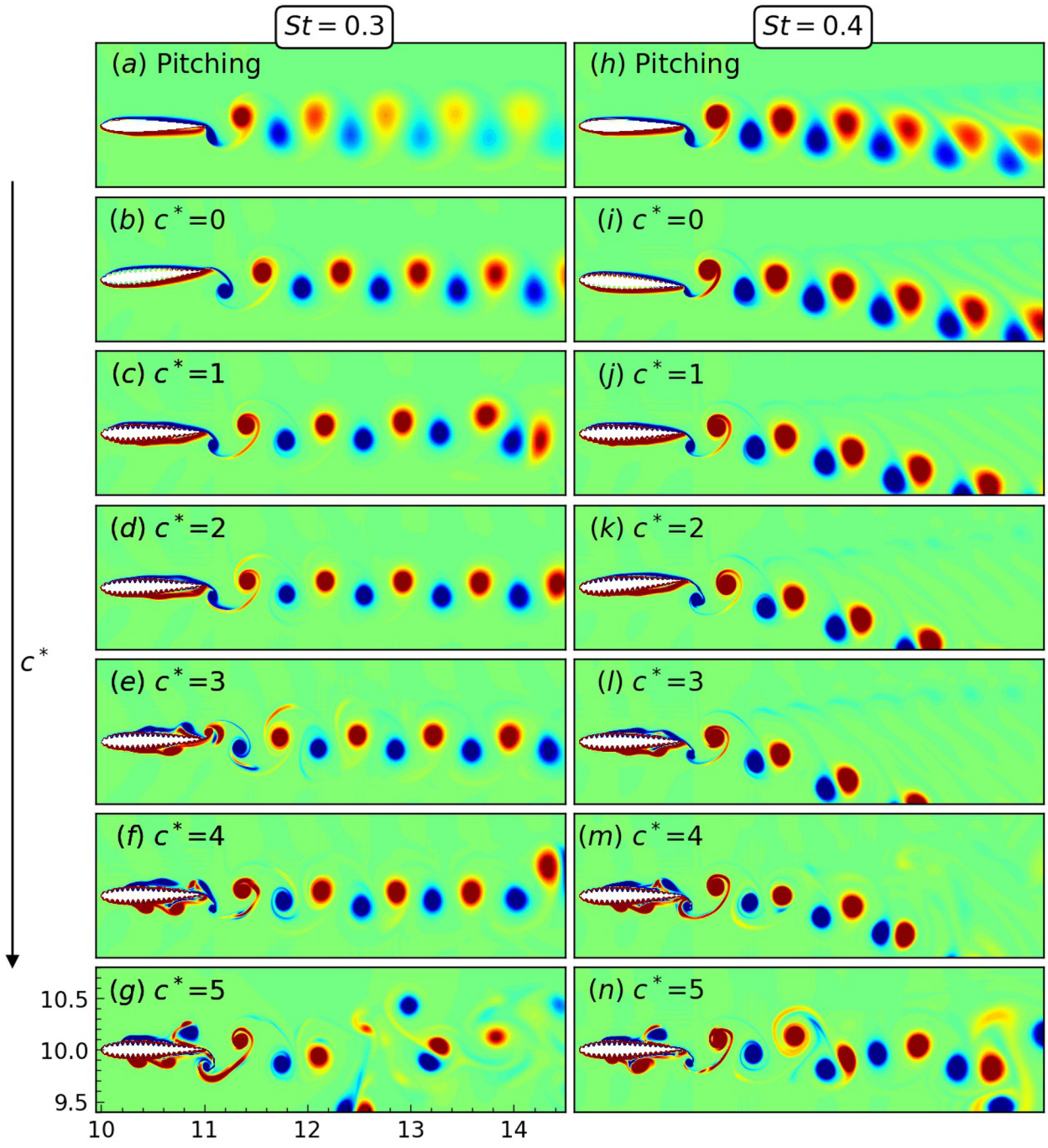


FIG. 5. Caption same as Fig. 4 for (a)–(g) $St = 0.3$ and (h)–(n) $St = 0.4$.

B. Characterization of the wakes

This section presents a detailed analysis of the various wake regimes discussed in Sec. III A. We analyze the average flow and self-induced velocity of vortex dipoles to characterize the wake.

1. Near-wake regimes

Figure 8 provides a schematic representation of the flow characteristics behind a hydrofoil subjected to pure surface undulation and combined motion. In the case of pure surface undulation at lower

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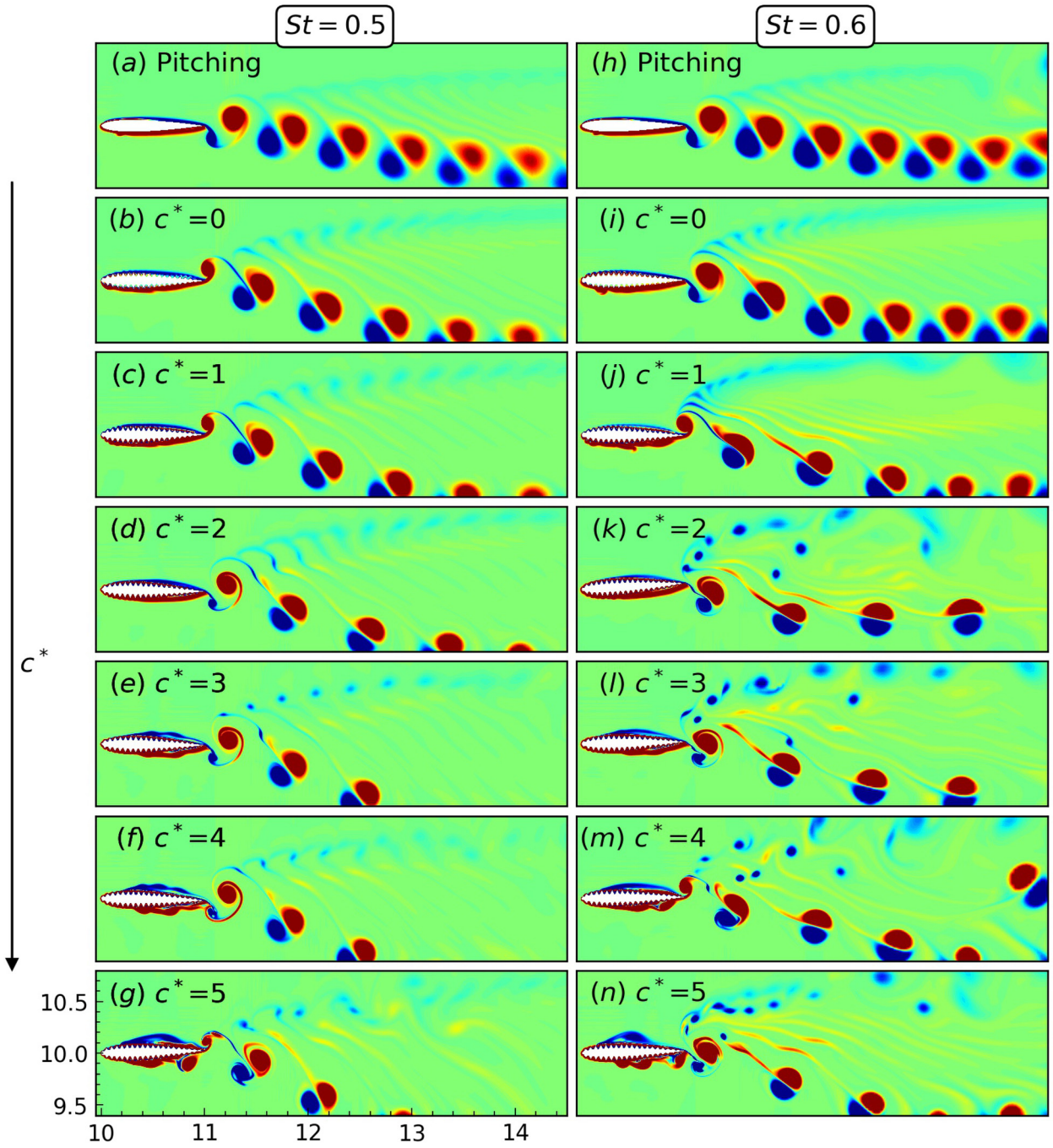


FIG. 6. Caption same as Fig. 4 for (a)–(g) $St = 0.5$ and (h)–(n) $St = 0.6$.

phase speeds ($c^* \leq 2$), flow separation occurs near the trailing edge, accompanied by the formation of a forward vortex sheet (FV_{Sheet}) in the vicinity of the trailing edge, as depicted in the schematic [Fig. 8(a)]. As the phase speed increases to $c^* \geq 3$, an additional vortex sheet is generated, as shown in Fig. 8(b). For the higher phase speed $c^* \geq 4$, the separation point moves upstream and closer to the leading edge. Consequently, the vortex sheets roll up and form a single vortex (side vortex) on each side of the hydrofoil, as shown in Fig. 8(b). The remaining counterclockwise (CCW) sheets above and clockwise (CW) sheets below lead to the formation of 2P-RV_{Sheet}.

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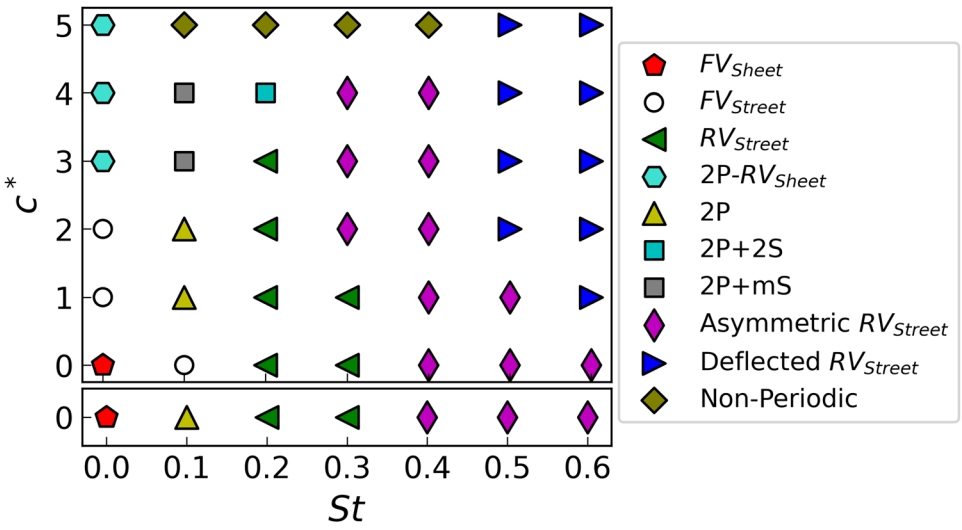


FIG. 7. Far wake regimes for the pure pitching ($c^* = 0$), pure surface undulation ($St = 0$), and the combined motion for different frequencies St (for pitching) and phase speeds c^* (for surface undulation). The bottom separate row shows the flow regimes for the pure pitching motion of a corrugated hydrofoil (also represented as $c^* = 0$). Symbols under the shaded regions are for cases involving pure pitching of corrugated hydrofoil.

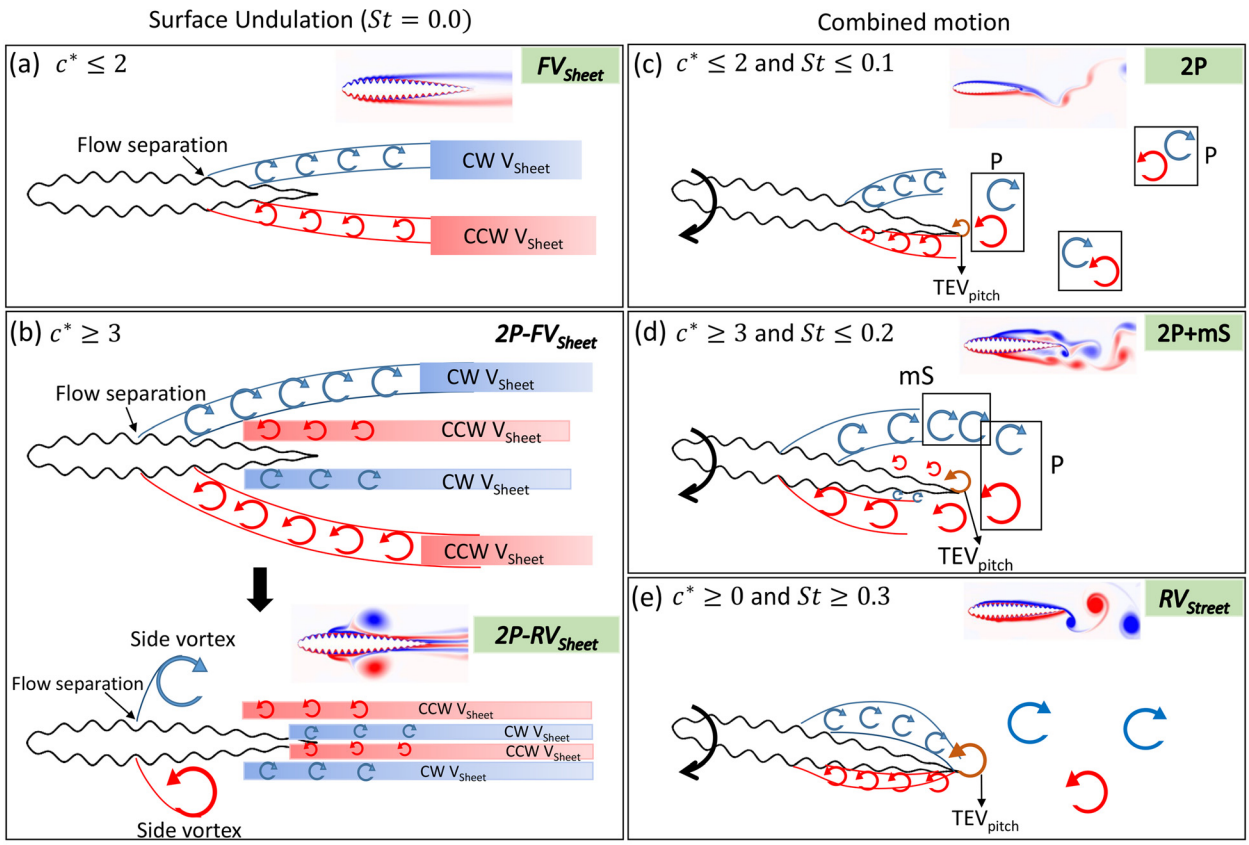


FIG. 8. Schematic representation of the vortex structures in the near wake. (a) and (b) The vortex structure for pure surface undulation with a wave number of $K = 20$, where the transition of 2P-FV_{Sheet} to 2P-RV_{Sheet} with the formation of side vortices (b). (c)–(e) The vortex structure for the combined motion of pitching with surface undulation. TEV_{pitch} represents a counterclockwise (CCW) vortex developed at the trailing edge due to the downward motion of the hydrofoil during pitching. The name of the wake is given in the top right of each panel, and representative vorticity contours for each panel are also plotted.

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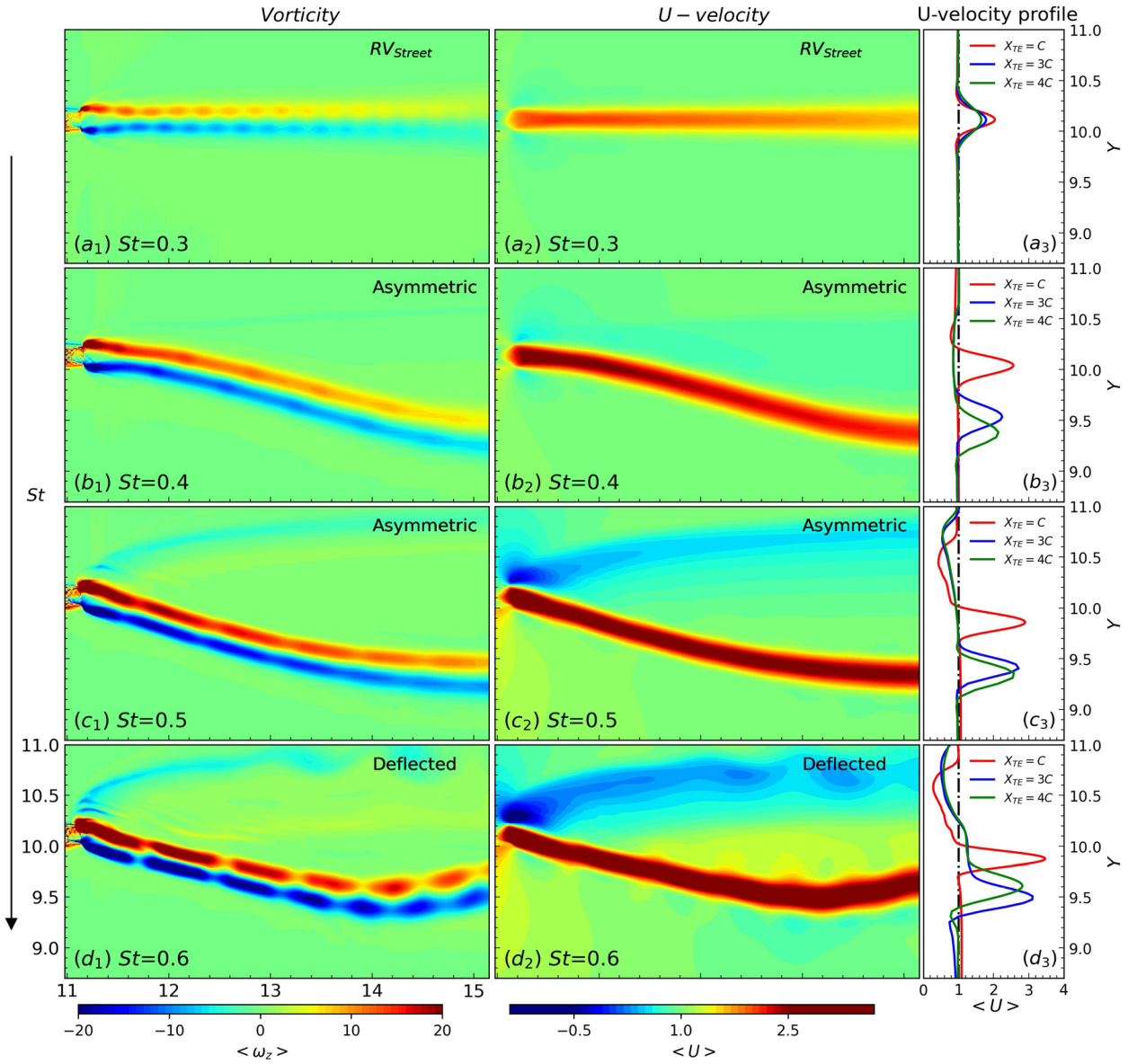


FIG. 9. Time averaged (a_1)–(d_1) vorticity contours, (a_2)–(d_2) streamwise U velocity contours, and (a_3)–(d_3) U-velocity profiles at a distance of C, $3C$, and $4C$ from trailing edge of the hydrofoil for various pitching frequency St at $c^* = 1$.

Figures 8(c)–8(e) show a schematic representation of the flow characteristics behind a hydrofoil subjected to the combined motion for different c^* and St. Under the combined motion, as the hydrofoil undergoes a downward pitching motion, counterclockwise (CCW) trailing-edge vortices (TEV_{pitch}) develop at the trailing edge [Fig. 8(c)]. The downward motion of the hydrofoil leads to the rolling up of the forward vortex sheet (FV_{Sheet}), separating a single vortex from both the sides. This separation gives rise to the formation of a pair of vortices (P) during a half cycle of pitching, while the trailing-edge vortices (TEV_{pitch}) merge and dissipate with the vortex sheets, as shown in Fig. 8(c). For surface undulation with higher phase speeds ($c^* \geq 3$), the clockwise (CW) vortex sheets divide into multiple single vortices when the hydrofoil pitches downward. Additionally, a counterclockwise (CCW) vortex sheet generates a single vortex, contributing to the formation of a pair of vortices (P) during a half cycle, accompanied by the generation of multiple vortices resulting from the disintegration of the vortex sheet, as shown in Fig. 8(d). Figure 8(e) schematically depicts the combined motion with $St \geq 0.3$ cases for all c^*. As St increases, the vortex sheets no longer spread along the transverse direction to the flow and converge toward the trailing edge instead. This behavior is attributed to stronger trailing-edge vortices induced by pitching, causing the vortex sheets to merge with these trailing-edge

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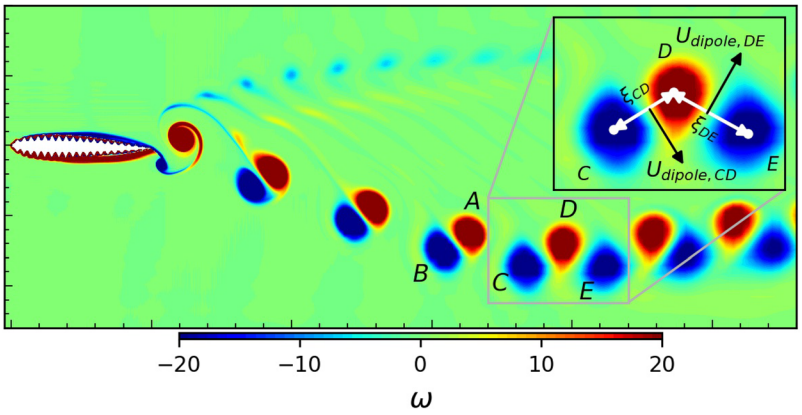


FIG. 10. Instantaneous vorticity contours for $St = 0.5$ and $c^* = 5$. Re-deflected asymmetric vortices can be seen due to the formation of vortex dipoles.

vortices. The merging of vortices further increases the vortex strength generated by the pitching motion. However, for $c^* = 5$ at $St = 0.3$ and 0.4, the flow does not show the formation of RV_{Street}; instead, a non-periodic flow exists behind the hydrofoil.

2. Far-wake regimes and characterization

Figure 9 depicts the cycle-averaged streamwise U-velocity contours and vorticity contours at $c^* = 1$ for various cases of St. The averaged contours indicate the presence of RV_{Street}, asymmetric RV_{Street}, and re-deflected RV_{Street} for $St = 0.3$, 0.4, and 0.6, respectively. The combined motion with pitching $St = 0.5$ also shows an asymmetric RV_{Street}. The asymmetric RV_{Street} is formed due to the vortex dipole formation, as explained by Godoy-Diana *et al.*[76] for the pure pitching motion of a hydrofoil. An increase in either St or c^* facilitates faster shedding of vortices from the hydrofoil surface, forming a vortex dipole with the advecting vortex behind the hydrofoil. The velocity profiles for three different locations are shown in Figs. 9(a3)–9(d3). At $St = 0.3$, the excess momentum remains almost at the same Y-position, as indicated by the U-velocity profiles, except for a decrease in the maximum value of U-velocity further downstream. This characteristic is attributed to the formation of RV_{Street} behind the hydrofoil. However, for higher St, specifically at $St = 0.4$ and $St = 0.5$, as the flow progresses downstream, the location of maximum velocity shifts downward in the vertical direction, reflecting the development of an asymmetric RV_{Street} [Figs. 9(b3) and 9(c3)].

On the other hand, at higher St ($St = 0.6$), the maximum streamwise velocity or jet exhibits a downward and upward vertical movement downstream, as shown in Fig. 9(d3). This trend suggests a re-deflection of the asymmetric RV_{Street} behind the hydrofoil. Additionally, a weak vortex street above the asymmetric RV_{Street} exists for the case of asymmetric RV_{Street} and re-deflected asymmetric RV_{Street}, which corresponds to the wake ($U < 1$) and jet region ($U > 1$) as shown by velocity profiles in Figs. 9(c3) and 9(d3). Overall, in the combined motion of surface undulation and pitching, the leading-edge vortices reinforce the trailing-edge vortices, increasing their strength and resulting in an early deflection of RV_{Street}. The re-deflection of the asymmetric RV_{Street} occurs in the opposite direction of the asymmetricity.

Further, we investigate the asymmetry and re-deflection of the reverse von Kármán vortex street (RV_{Street}) for the combined motion.

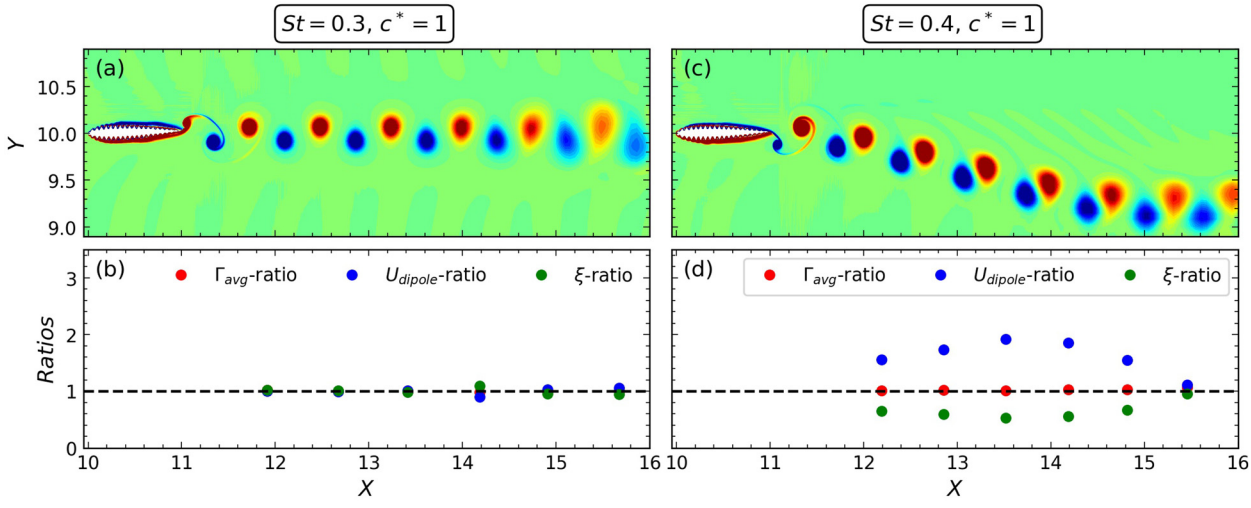


FIG. 11. (a) and (c) Instantaneous vorticity contours and (b) and (d) ratios of vorticity dipoles in the streamwise direction for $St = 0.3$ and $St = 0.4$ at $c^* = 1$.

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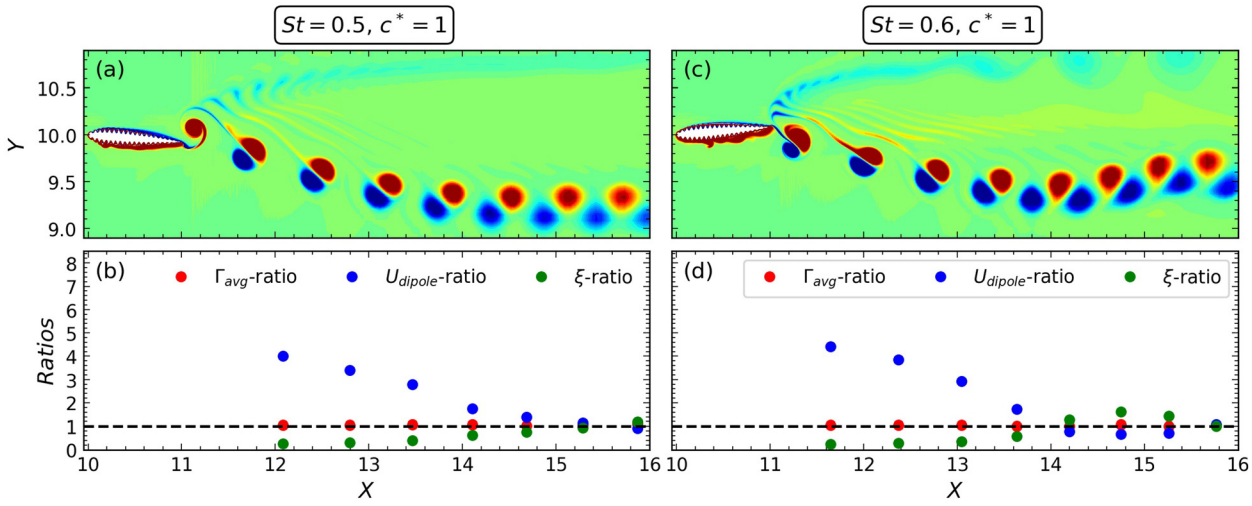


FIG. 12. (a) and (c) Instantaneous vorticity contours and (b) and (d) ratios of vorticity dipoles in the streamwise direction for $St = 0.5$ and $St = 0.6$ at $c^* = 1$.

Similar wake changes were also reported in other propulsive motions. For example, Lewin and Haj-Hariri[77] demonstrated that in the case of pure heaving motion, the vorticity at the leading edge of a hydrofoil contributes to either reinforcing the trailing-edge vortices or pairing up with them. Similar findings were reported for combined pitching and heaving cases in the study of Patel and Bhardwaj.[8] Gupta *et al.*[11] also reported an asymmetricity in RV_{Street} for an undulating hydrofoil.

Next, we discuss the vortex dynamics of the wake for the case shown in Fig. 10. The formation of a vortex dipole with vortices *A* and *B* is illustrated in Fig. 10, with the leading vortex *A* being a counter-clockwise vortex and the dipole moving downward with the vortex dipole velocity. At a further downstream location, due to advection, the vortices *C* and *E* approach each other and form a different dipoles with vortex switching, as explained by Wei and Zheng[78] for a heaving hydrofoil. The vortex *E*, being the leading vortex in the dipole and a clockwise vortex, results in the re-deflection of the asymmetric RV_{Street} in the opposite direction of the asymmetricity.

The competition between two sets of vortex dipoles, namely, the symmetry-breaking dipole (*C* and *D*) and the symmetry-holding dipole (*D* and *E*), results in the development of flow asymmetry and re-deflection behind the hydrofoil. This phenomenon was extensively investigated by Godoy-Diana *et al.*[76] and Wei and Zheng.[78] The interaction between these vortex dipoles can be quantified using self-induced dipole velocity U_{dipole} (Fig. 10), expressed as follows:

$$U_{\text{dipole}} = \frac{\Gamma_{\text{avg}}}{2\pi\xi}, \tag{12}$$

where ξ represents the center-to-center distance between the two vortices in the vortex dipole as shown in the inset of Fig. 10, and Γ_{avg} is the average absolute circulation of the two vortices in the dipole. In

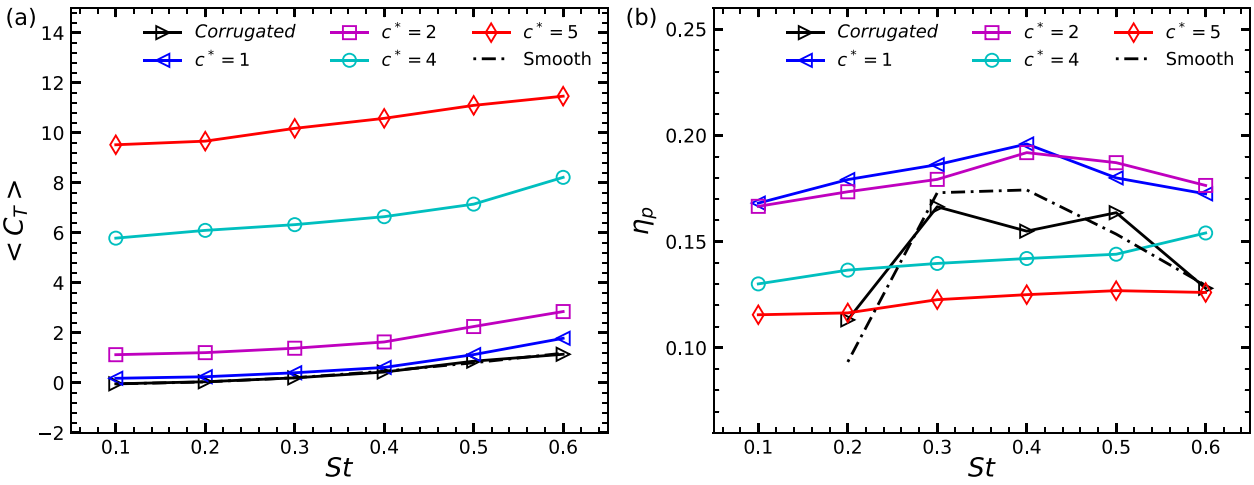


FIG. 13. Variation of (a) mean coefficient of thrust $\langle C_T \rangle$ and (b) propulsive efficiency η_p with increasing pitching Strouhal number *St* for various surface undulation phase speed c^*. The dotted line shows the variation of $\langle C_T \rangle$ and η_p for the case of only pitching motion.

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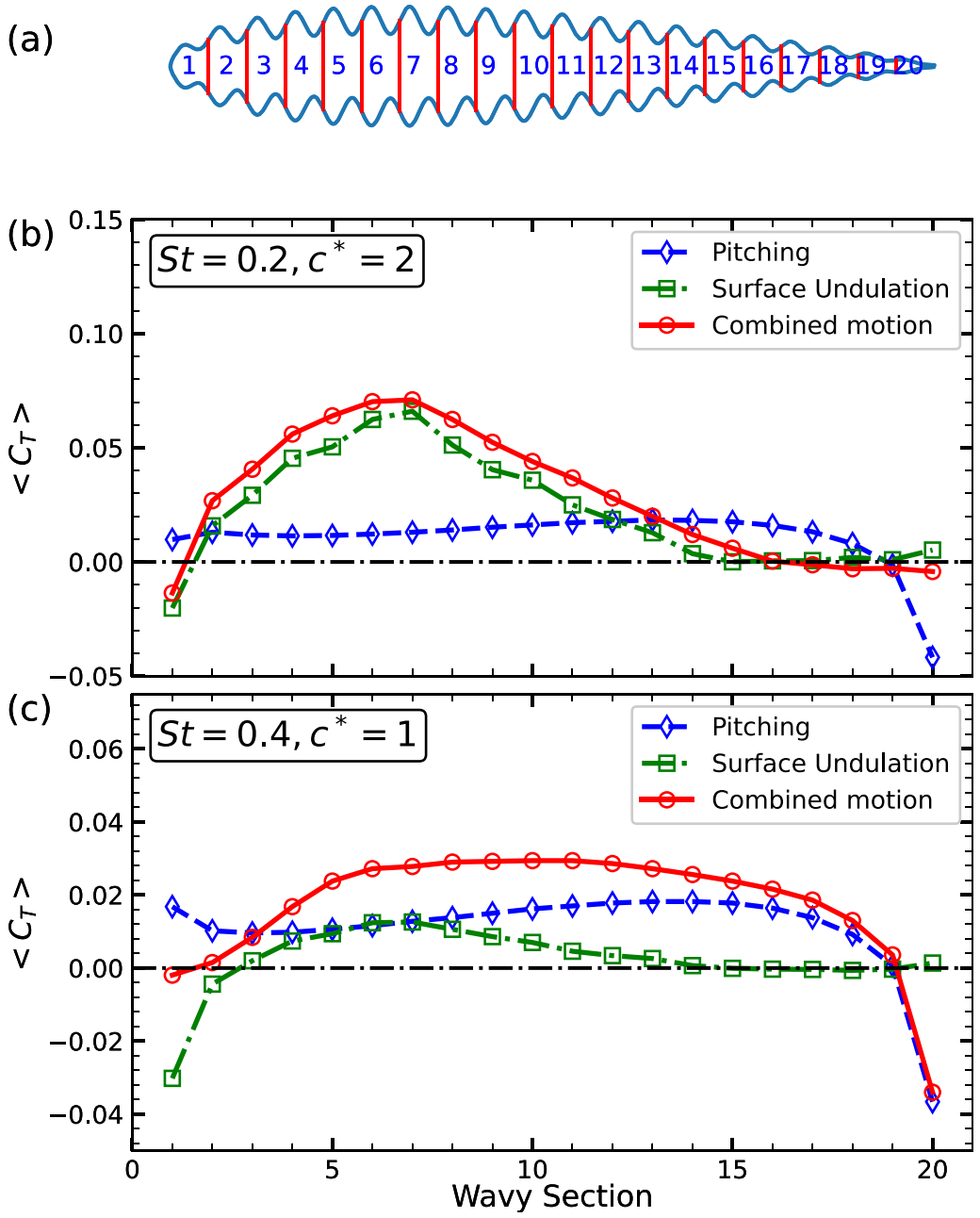


FIG. 14. (a) Hydrofoil with 20 equal lengths of time-wise varying wavy sections. Time-averaged thrust coefficient variation along the wavy sections for pure pitching, pure surface undulation, and combined motion is presented at (b) $St = 0.2$ and $c^* = 2$ and (c) $St = 0.4$ and $c^* = 1$.

our analysis, we have computed U_{dipole} for both the symmetry-holding and the symmetry-breaking dipoles based on temporal vorticity contours downstream of the hydrofoil, considering different vorticities.

Motivated by Shah *et al.*,[79] we calculate several ratios for the vortex dipoles to investigate the behavior of asymmetry and re-deflection in the RV_{Street} flow behind the hydrofoil. Specifically, we examine the dipole speed ratio (U_{dipole}-ratio), the average circulation ratio (Γ_{avg}-ratio), and the distance ratio (ξ-ratio), defined as follows:

$$U_{\text{dipole}} - \text{ratio} = \frac{U_{\text{dipole,CD}}}{U_{\text{dipole,DE}}}, \quad \Gamma_{\text{avg}}\text{-ratio} = \frac{\Gamma_{\text{avg, CD}}}{\Gamma_{\text{avg,DE}}}, \quad \xi\text{-ratio} = \frac{\xi_{\text{CD}}}{\xi_{\text{DE}}}, \tag{13}$$

where subscripts *CD* and *DE* represent symmetry-breaking and symmetry-holding vortex dipoles, respectively, and do not indicate any

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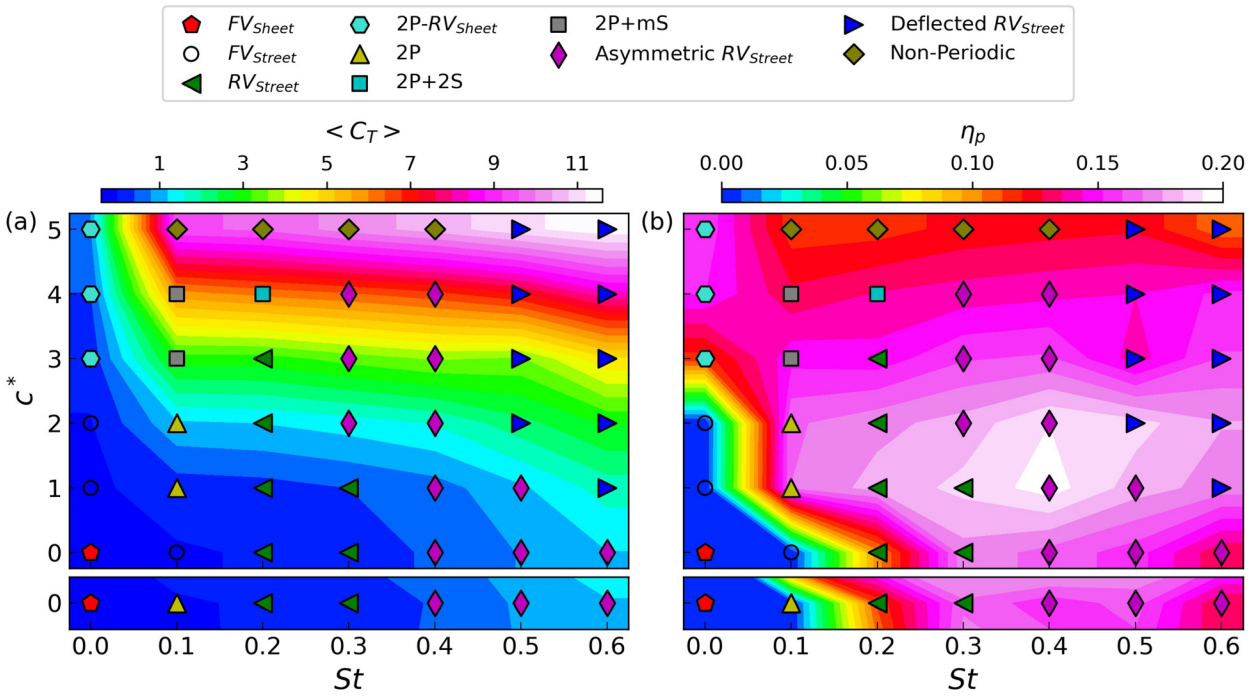


FIG. 15. Regime maps based on the various flow modes behind the hydrofoil are shown for various pitching St and c^*. The maps are overlapped on contours of (a) time-averaged coefficient of thrust $\langle C_T \rangle$ and (b) propulsive efficiency η_p. The pure pitching of smooth and corrugated hydrofoil is presented by $c^* = 0$. The bottom subplot shows the case of pure pitching of a corrugated hydrofoil.

specific location downstream of the hydrofoil. The U_{dipole} and ξ are schematically represented in Fig. 10.

Figures 11 and 12 illustrate the temporal vorticity contours and different ratios defined in Eq. (13), for the case of different St at $c^* = 1$. These ratios are computed for different combinations of dipoles downstream of the hydrofoil with the combined motion. The detailed calculation procedures of these ratios are given in Subsection 1 of Appendix. In Fig. 11(b), for $St = 0.3$, the ratios remain relatively constant and close to unity. This indicates that the distance between the dipoles and their average circulations is nearly identical for the symmetry-breaking and symmetry-holding vortex dipoles. Consequently, the self-induced dipole velocity (U_{dipole}) is approximately the same for both the vortex dipoles, leading to a symmetrical formation of the von Kármán vortex street (RV_{Street}) behind the hydrofoil, as depicted in Fig. 11(a). U_{dipole}-ratio slightly below 1 signifies that the vortex dipole responsible for maintaining symmetry, specifically vortices D and E, exhibits slightly larger strength than the vortex dipole responsible for the symmetry-breaking vortex dipole (C and D).

In the case of $St = 0.4$ at $c^* = 1$, the U_{dipole}-ratio increases due to a reduction in the distance ratio, ξ-ratio, between the vortex dipoles as we move downstream, as shown in Fig. 11(d). This increase in U_{dipole}-ratio signifies the asymmetry in RV_{Street}. Further downstream, the distance ratio between the dipoles increases while the U_{dipole}-ratio decreases. This behavior causes the vortex street downstream to align parallel to the horizontal mean position of the hydrofoil with the combined motion. Similarly, in Fig. 12(b), the distance ratio near the trailing edge of the hydrofoil is minimal, indicating a large distance between the symmetry-holding dipole vortices. Away from the trailing edge downstream, the distance ratio increases. In contrast, U_{dipole}-ratio decreases and approaches closer to 1, indicating that the vortex street becomes parallel to the mean position of the hydrofoil downstream.

For $St = 0.6$ at $c^* = 1$, a similar trend is observed near the trailing edge of the hydrofoil as in the case of $St = 0.5$ at $c^* = 1$. However, in downstream, the U_{dipole}-ratio starts to decrease and goes below unity, indicating a re-deflection of the asymmetric RV_{Street}, as shown in Figs. 12(c) and 12(d). The sub-unity value for the U_{dipole}-ratio indicates that the symmetry-holding vortex dipole (D and E) is stronger than the symmetry-breaking dipole (C and D). It is important to note that the circulation ratio, denoted as Γ_{avg}-ratio, exhibits a consistent behavior for all downstream vortex dipole couples in the range of $St = 0.3$ to $St = 0.6$ at $c^* = 1$, as depicted in Figs. 11 and 12. In each case, the Γ_{avg}-ratio remains close to 1, indicating a near balance in the average circulations of the vortex dipoles for symmetry breaking and symmetry holding dipoles.

C. Propulsive performance

Figure 13 displays the time-averaged values of thrust coefficient and propulsive efficiency for various pitching Strouhal numbers St plotted against different surface undulation phase speeds c^*. The mean coefficient of thrust $\langle C_T \rangle$ and propulsive efficiency η_p for the case of pure pitching motion are depicted by the dashed-dot black line in Figs. 13(a) and 13(b), respectively. The results demonstrate that $\langle C_T \rangle$ increases with an increase in both c^* and St, implying that incorporating the traveling wave-based surface undulation with pitching motion enhances the thrust generation on the hydrofoil. The combined effect

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of localized jet formation in the wave trough and the reverse vortex Kármán street in the wake of the hydrofoil contribute to the increase in thrust force with the combined motion. Figure 13(b) depicts the variation of propulsive efficiency, demonstrating that combining surface undulation with pitching motion increases efficiency for all pitching St compared to the case of pure pitching motion for a particular phase speed. The maximum efficiency is observed for the combined motion at $c^* = 1$ for $St = 0.4$.

Figure 14 exhibits a spatial variation of the thrust coefficient along the chord length of the hydrofoil, wherein the hydrofoil is segmented into 20 equal-length distinct wavy sections based on the wave number $K = 20$, as represented through schematic in Fig. 14(a). The thrust variation of combined motion along the chord length is compared with pure pitching and pure surface undulation motion. The spatial variation of the time-averaged thrust coefficient in Fig. 14(b) shows that for the combined motion and pure surface undulation, it follows an increasing–decreasing trend till wavy section 19. However, for wavy section 20, the combined motion generates drag, whereas pure surface undulation results in thrust. In contrast, pure pitching motion generates a constant thrust across all wavy sections, except for wavy section 20, where drag is generated. Note that for the combined motion, wavy section 1 and wavy section 20 generate drag, while all other wavy section generates a time-averaged thrust for a cycle. The time-averaged thrust for the combined motion follows a similar trend as that of the pure surface undulation case, except for wavy section 20, which resembles the pure pitching motion trend.

We plot the spatial variation of time-averaged thrust along the wavy sections for the case of maximum efficiency ($c^* = 1, St = 0.4$) for the combined motion in Fig. 14(c). Notably, the combined motion is producing almost constant thrust forces for the wavy sections 6–12. For wavy sections 2–5, where the pure pitching motion shows a dip in the thrust coefficient, $\langle C_T \rangle$ of the combined motion is compensated by the pure surface undulation motion. The trend of the spatial variation of the time-averaged thrust coefficient for different wavy sections for the combined motion closely resembles the trend with the addition of the time-averaged thrust coefficient for pure traveling wave motion and pure pitching motion, as shown in Figs. 14(b) and 14(c).

D. Variation of wake with propulsive performance

As described earlier, applying traveling wave-based surface undulation on a pitching hydrofoil produces nine distinct vortex sheet/street patterns. A non-periodic flow regime is also observed at lower pitching frequencies and higher surface undulating phase speeds. The vortex patterns for various combinations of pitching St and surface undulation c^* are presented in Fig. 15, along with contours for the thrust coefficient and propulsive efficiency shown in Figs. 15(a) and 15(b), respectively.

Analysis of the flow regimes and propulsive parameters reveals that the maximum thrust is achieved with a higher c^* and St combination for the re-deflected RV_{Street} pattern. In comparison, the maximum

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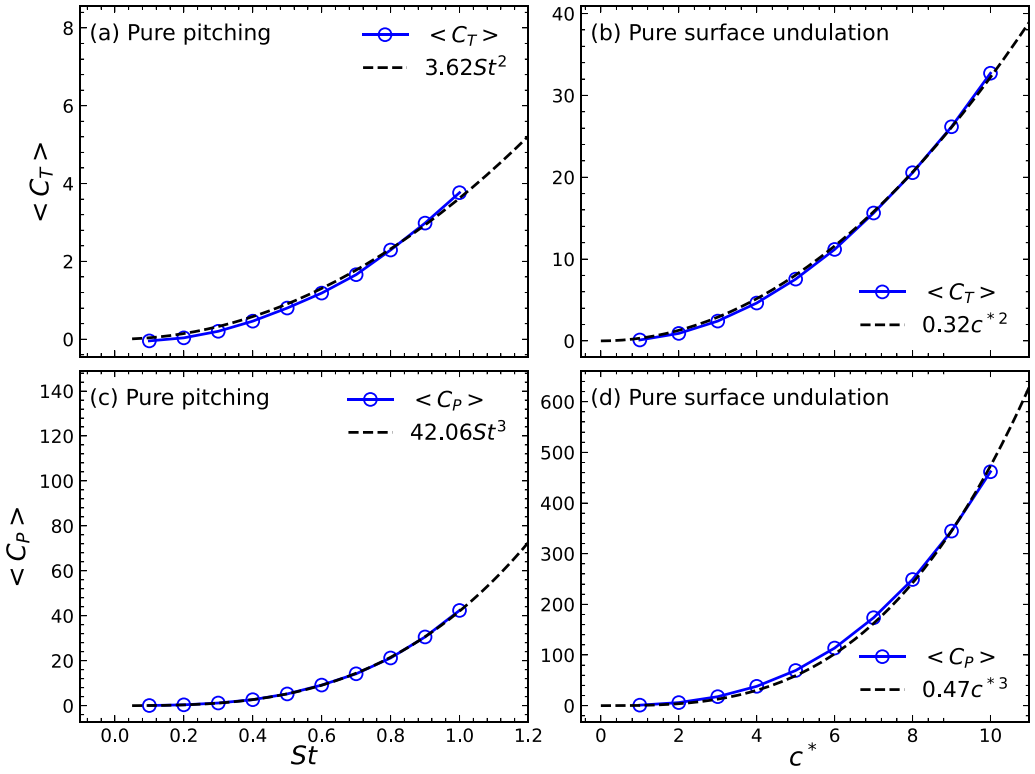


FIG. 16. Curve fitting of thrust coefficient for (a) and (c) only pitching case and (b) and (d) only traveling wave case is presented. (a) and (b) The curve fitting of the thrust coefficient, and (c) and (d) the same power coefficient.

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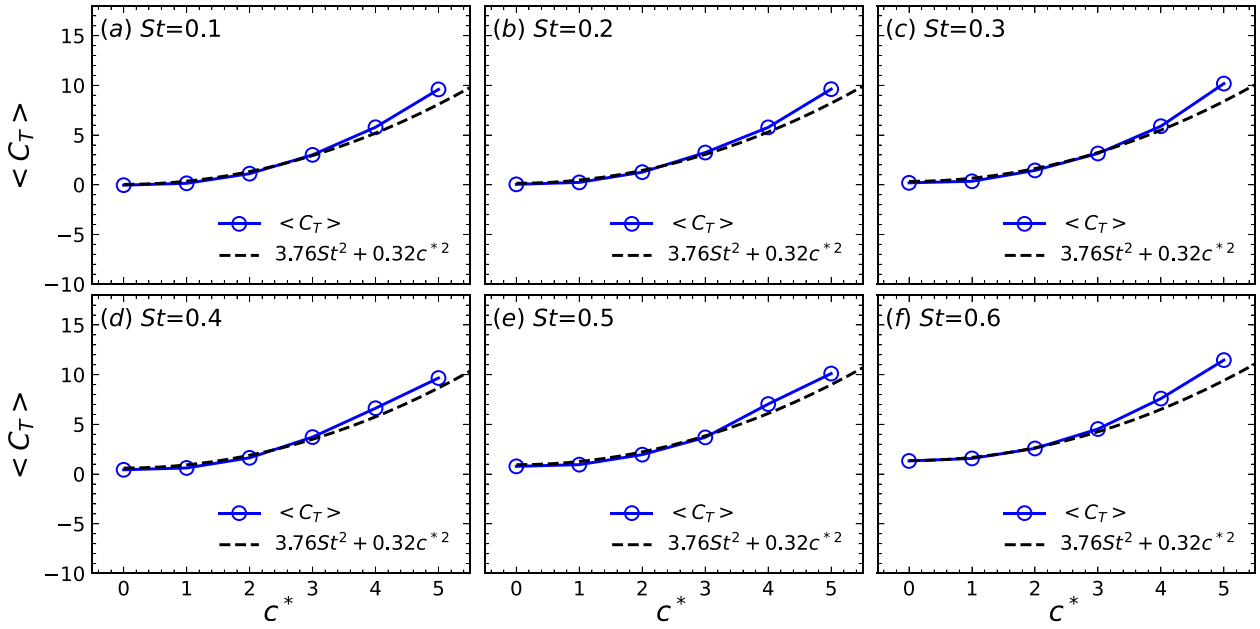


FIG. 17. Thrust coefficient is plotted along with the scaling curve for (a) $St = 0.1$, (b) $St = 0.2$, (c) $St = 0.3$, (d) $St = 0.4$, (e) $St = 0.5$, and (f) $St = 0.6$.

efficiency is achieved with an intermediate c^* and St combination for an asymmetric RV_{Street}.

E. Scaling analysis

We have developed a scaling law for $\langle C_T \rangle$ and input power coefficient, $\langle C_P \rangle$. In a previous study, Das *et al.*[22] have given a scaling relation for a hydrofoil with pitching motion. It has been found in several previous works that $\langle C_T \rangle \sim St^2$ and $\langle C_P \rangle \sim St^3$.[22,30,32,80] Floryan *et al.*[30] developed a scaling law by considering added mass forces in the case of a purely pitching hydrofoil. Interestingly, they also observed the same scaling behavior for $\langle C_T \rangle$ as Das *et al.*[22] did, provided that the pitching amplitude and drag on the hydrofoil remained small.

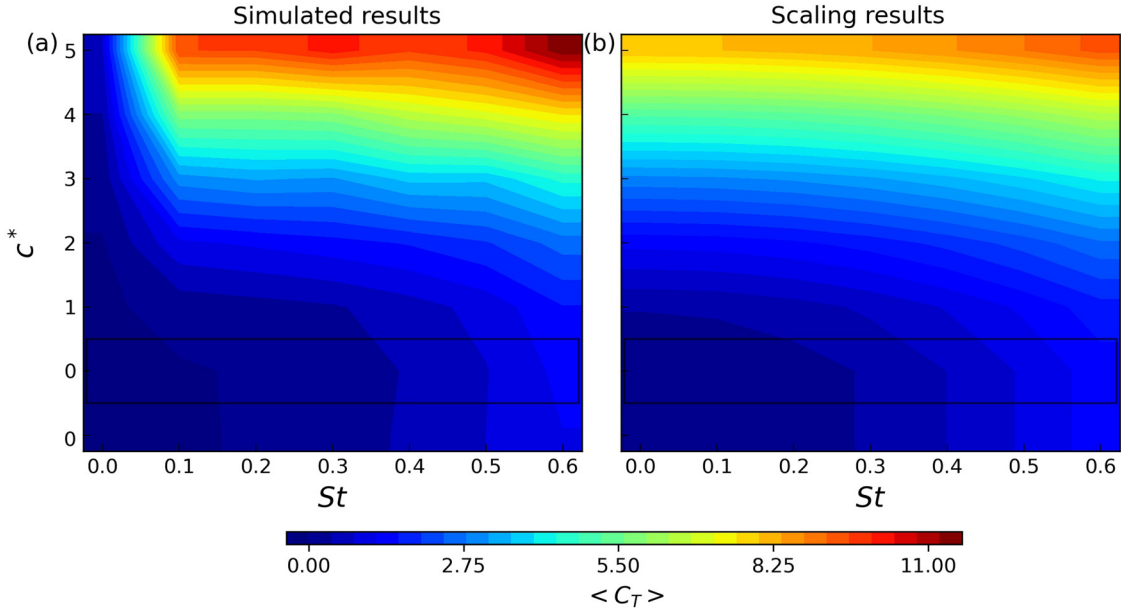


FIG. 18. Thrust coefficient contours at different pitching St and surface undulation c^* are presented for (a) simulated results and (b) scaling results. The pure pitching of smooth and corrugated hydrofoil is presented by $c^* = 0$. The rectangular box shows the case of pure pitching of a corrugated hydrofoil.

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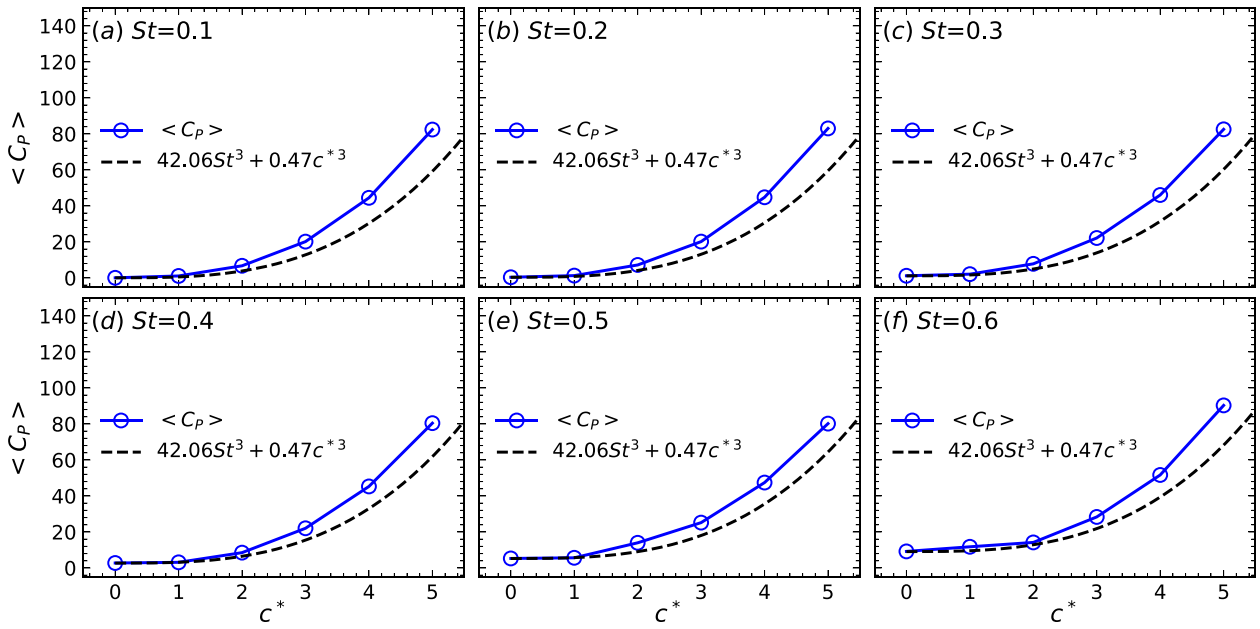


FIG. 19. Time-averaged power coefficient is plotted along with the scaling curve for (a) $St = 0.1$, (b) $St = 0.2$, (c) $St = 0.3$, (d) $St = 0.4$, (e) $St = 0.5$, and (f) $St = 0.6$.

In our previous work,[14] we found that $\langle C_T \rangle \sim c^{*2}$, while $\langle C_P \rangle \sim c^{*3}$. Therefore, for combined motion, we have linearly superimposed these scaling laws for thrust coefficient and power coefficient as $\langle C_T \rangle \sim C_1 St^2 + C_2 c^{*2}$ and $\langle C_P \rangle \sim C_3 St^3 + C_4 c^{*3}$, respectively. The values of C_1 and C_3 are determined through curve fitting, using pure pitching case data, while C_2 and C_4 are calculated using only traveling wave-based surface undulation data as shown in Fig. 16. Therefore, the scaling laws can be superimposed for the combined motion and can be expressed as

$$\langle C_T \rangle \sim 3.76 St^2 + 0.32 c^{*2}, \tag{14}$$

$$\langle C_P \rangle \sim 42.06 St^3 + 0.47 c^{*3}. \tag{15}$$

Figures 16(a) and 16(b) show the curve fitting of $\langle C_T \rangle$ for pure pitching and pure traveling wave case, respectively. The simulated data for $\langle C_T \rangle$ are presented along with the scaling data in Fig. 17, for different St and c^*. It perfectly fits the scaling law, except for the higher c^*. Figure 18 compares the contour of the thrust coefficient obtained from the simulated data and the thrust predicted by the proposed scaling law based on the linear superimposition. The results reveal a remarkable agreement between the scaling predictions and the simulated thrust generation. The contours derived from the scaling law closely match the simulated results, indicating the effectiveness of the proposed approach in accurately estimating thrust. Figure 19 also shows a good match with the linearly superimposed scaling model. However, the scaling model underestimates the input power for higher c^*.

IV. DISCUSSION AND CONCLUSIONS

The propulsive performance of a combined motion of pitching with surface traveling wave on a hydrofoil is investigated numerically here using an in-house level set-based sharp interface immersed boundary code. The mean-to-peak movement for pitching has been fixed at 5°. For traveling wave-based surface undulation, the non-dimensional amplitude and the wave number have been fixed at 0.15 and 20, respectively. c^* of traveling wave and St of pitching are the only parameters that have been varied from 1 to 5 and 0.1 to 0.6, respectively. The conclusions drawn from the present study are as follows:

1. The flow behind hydrofoil at very low $St = 0.1$ changes from a von Kármán street for pure pitching case to a 2P vortex street for the combined motion as we increase the surface undulation phase speed $c^* \leq 2$. Further increases in c^* lead to the formation of 2P + mS vortex street. This transition is attributed to the disintegration of the vortex sheets generated by surface undulation due to the rolling effect induced by the pitching motion, disrupting what would otherwise be relatively steady vortex sheets. Notably, this phenomenon occurs at relatively low Strouhal numbers ($St \leq 0.2$).
2. The generation of side vortices for the pure surface undulation is initiated by the movement of flow separation points toward the leading edge, causing the vortex sheet to roll up and leading to the development of bigger side vortices. Additionally, the presence of flow reversal at the crests of the traveling waves contributes to the reinforcement of these side vortices. These side vortices form at higher surface undulation phase speed ($c^* \geq 3$).
3. The asymmetricity of reverse von Kármán vortices about the centerline of hydrofoil in the wake is observed even at lower St with the combined motion in comparison to pure pitching.
4. At higher pitching St, the asymmetric RV_{Street} gets re-deflected. We quantified the asymmetry and re-deflection in the reverse von Kármán vortex street (RV_{Street}) by assessing various ratios between the symmetry-breaking and symmetry-holding vortex dipoles formed behind the hydrofoil. The presence of asymmetry in the RV_{Street} is linked to the dominance of the symmetry-breaking vortex dipole.

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Conversely, the re-deflection of asymmetric vortices is attributed to the prevailing influence of the symmetry-holding vortex dipole.

5. The combined motion of pitching with the traveling wave contributes to an increase in the mean thrust coefficient due to the combined effect of reverse von Kármán street in the wake and the localized velocity jet in the trough of the wave. $\langle C_T \rangle$ increases with an increase in both St and c^*. The mean thrust generated is always more than that of the pitching-only case. The propulsive efficiency η_p is the maximum where the pitching and traveling wave motion compete with each other in thrust generation. The propulsive efficiency has been observed to be maximum at lower $c^* = 1$ for $St = 0.4$.

The present results signify the effect of surface undulation in increasing the efficiency of a pitching hydrofoil. The result highlights that the competition between pitching and surface undulation creates an efficient combined motion for thrust generation. The present bio-inspired research holds the potential to inspire the design and development of specialized autonomous underwater vehicles tailored to specific needs.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Sarvesh Shukla: Conceptualization; Formal analysis; Investigation; Writing – original draft; Writing – review & editing. **Atul Sharma:** Conceptualization; Formal analysis; Investigation; Supervision; Writing – review & editing. **Amit Agrawal**: Conceptualization; Formal analysis; Investigation; Supervision; Writing – review & editing. **Rajneesh Bhardwaj**: Conceptualization; Formal analysis; Investigation; Supervision; Writing – review & editing.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

NOMENCLATURE

A_L	Local amplitude-to-thickness ratio (a_L/h)
A_p	Non-dimensional mean to maximum TE lateral displacement (a_p/L)
a_L	Local amplitude of traveling wave (m)
a_p	Mean to a maximum lateral TE displacement of pitching hydrofoil
C	Chord length of hydrofoil (m)
C_L	Coefficient of lift
C_P	Input power coefficient
C_T	Coefficient of thrust $(2F_T/\rho U_\infty{}^2 L)$
c	Wave speed of traveling wave based surface undulation (ms^{-1})
c^*	Phase speed of surface undulation (c/U_∞)
F_L	Lift force over whole hydrofoil (N)
F_T	Thrust force (N)
f	Traveling wave frequency (s^{-1})
f_p	Pitching frequency (s^{-1})
f_L	Local lift force (N)
K	Wave number of traveling wave (L/λ)
P_{in}	Input power
Re	Reynolds number $(\rho U_\infty C/\mu)$
St	Strouhal number of pitching $(2f_p a_p/U_\infty)$
T	Dimensionless time span for time-averaging
t	Time (s)
U	Dimensionless local flow velocity
U_∞	Free-stream velocity (ms^{-1})
U_c	Dimensionless longitudinal velocity of hydrofoil surface (u_c/U_∞)
u_c	Longitudinal velocity of hydrofoil surface (ms^{-1})
V_c	Dimensionless lateral velocity of hydrofoil surface (v_c/U_∞)
v_c	Lateral velocity of hydrofoil surface (ms^{-1})
X_c	Dimensionless X-coordinates of the combined motion
Y_c	Dimensionless Y-coordinates of the combined motion
Y_{s0}	Dimensionless Y-coordinates of undeformed hydrofoil profile
y_{s0}	Y-coordinates of undeformed hydrofoil profile (m)
$\langle\rangle$	Time-averaged quantity

Greek symbols

μ	Dynamic viscosity $(\mathrm{N\,m^{-2}s})$
λ	Wavelength of surface undulation wave
θ	Temporal angular displacement of pitching
θ_0	Pitching amplitude
ρ	Density $(\mathrm{kg\,m^{-3}})$
τ	Dimensionless time (tU_∞/C)
ξ	Distance between two vortices in the dipole
Γ	Circulation of vortex

Subscripts

avg	Average
$dipole$	Dipole
rms	Root mean square value

Acronyms

FV_{Sheet}	Forward vortex sheet
FV_{Street}	Forward vortex street
LS-IIM	Level set-based immersed interface method
RV_{Sheet}	Reverse vortex sheet
RV_{Street}	Reverse vortex street

APPENDIX: CHARACTERIZATION OF VORTEX DEFLECTION

1. Computation of ξ

To determine the positions of a vortex center within any pair of vortices with opposite signs, we employed a method involving the use of arbitrary sized rectangular enclosures that

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encapsulate the vortices entirely. Subsequently, we identified the centers of these vortices by locating the local maxima and minima within the positive (anti-clockwise) and negative (clockwise) vortices, respectively. Following this identification of centers of the vortices, we calculated the distance between these two vortex centers, denoted for vortices C and D in Fig. 10, using the following formula:

$$\xi_{CD} = \sqrt{(x_C - x_D)^2 + (y_C - y_D)^2}. \tag{A1}$$

This procedure elucidates the methodology for locating and quantifying the distance between vortex centers ξ.

2. Computation procedure for circulation Γ

To compute the circulation of a vortex, we employ a systematic approach. Initially, we define a rectangular region of arbitrary size that fully encompasses the entire vortex under consideration. Within this region, we analyze the vorticity data and identify a region where the vorticity is greater than 5% of the maximum vorticity value, similar to the Kumar *et al.*[81] Subsequently, we determine the boundary of these selected points by employing the Quickhull algorithm, resulting in the convex hull boundary. Figure 20 visually illustrates the selected rectangular area and the corresponding convex hull boundary for the vortex. Once we have obtained these boundary points, we proceed to compute the vortex circulation using the line integral method, which is expressed as

$$\Gamma = \oint \mathrm{U} \cdot \boldsymbol{d}\mathbf{r}. \tag{A2}$$

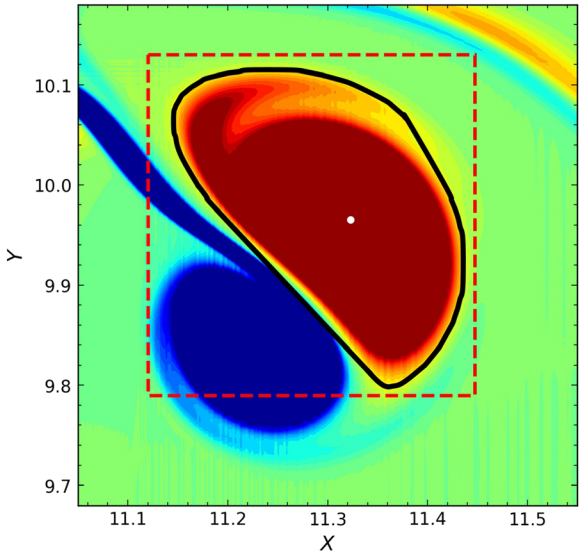


FIG. 20. Rectangular zone, delineated by red dotted lines, covering the entire vortex with vorticity contour for the $St = 0.6$ and $c^* = 1$ case, and the convex hull boundary indicated by the black line. Here, the white dot shows the center of the vortex.

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